

THEORIES OF THE DISTRIBUTION OF EARNINGS *

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Abstract

Several empirical regularities motivate most theories of the distribution of labor earnings. Earnings distributions tend to be skewed to the right and display long right tails. Mean earnings always exceed median earnings and the top percentiles of earners account for quite a disproportionate share of total earnings. Mean earnings also differ greatly across groups defined by occupation, education, experience, and other observed traits. With respect to the evolution of the distribution of earnings for a given cohort, initial earnings dispersion is smaller than the dispersion observed in prime working years.

We explore several models that address these stylized facts. Stochastic theories examine links between assumptions about the distribution of endowments and implied features of earnings distributions given assumptions about the processes that translate endowments into earnings. Selection models describe how workers choose a career. Because workers select their best option from a menu of possible careers, their allocation decisions tend to generate skewed earnings distributions. Sorting models illustrate this process in an environment where workers learn about their endowments and therefore adjust their allocation decisions over time.

Human capital theory demonstrates that earnings dispersion is a prerequisite for significant skill investments. Without earnings dispersion, workers would not willingly make the investments necessary for high-skill jobs. Human capital models illustrate how endowments of wealth and talent influence the investment decisions that generate observed distributions of earnings.

Agency models illustrate how wage structures may determine rather than reflect worker productivity. Tournament theory addresses the long right tails of wage distributions within firms. Efficiency wage models address differences in wages across employments that involve different monitoring technologies.

Keywords: Human capital, selection, sorting, agency, comparative advantage, skewness

JEL codes. J2, J3

0. Introduction

Theories of earnings distributions have always had strong empirical motivations. The earliest empirical studies of wealth and income discovered a remarkable regularity that is found in all observed earnings distributions in large populations. Earnings distributions (and income distributions more generally) are always skewed to the right. Their density functions are asymmetric and display a long right tail and positive skewness measure (third moment about the mean). They are also leptokurtic (positive fourth cumulant) and have a “fat tail”. Put differently, mean earnings always exceed median earnings. The top percentiles of earners account for a strikingly disproportionate share of total earnings.

Pareto (1897) and Bowley (1915) pioneered the assembly and empirical analysis of data on personal wealth and incomes. Later, extensive development of microeconomic data sources led to a more comprehensive development of the subject. Early work attempted to formulate the economic and statistical basis for fitting specific functional forms to empirical distributions. Were earnings capacities log-normally distributed, Paretian, or something else (Staele, 1943; Miller, 1955; Lebergott, 1959; Harrison, 1981)?

Various theories of the distribution of earnings address not only the fat tail of earnings distributions but also several other empirical observations. Earnings differ greatly across groups of workers defined by occupation, education, experience, and other observed traits. Earnings also vary within groups that are observationally similar. Further, earnings dispersion for a particular cohort of workers is greater among experienced workers than among workers who are beginning their careers. In a given cohort, much of the eventual inequality in lifetime earnings is not apparent until the workers are well into their careers.

Below, we describe four types of models that offer explanations for some or all of the observations. Stochastic theories begin with distributional assumptions about worker endowments and examine what kinds of stochastic structures might be consistent with observed earnings distributions. Selection models also begin with assumptions about the distribution of endowments but worker decisions concerning how to allocate their skills to specific tasks generate the overall distribution of earnings. Sorting models describe how workers change these allocations as they learn more about their endowments. Sorting contributes to the evolution of the distribution of earnings for a particular cohort. Human capital theory describes how workers acquire skills. It demonstrates that earnings inequality is a necessity in an economy where some activities require more costly investments than others. These models also illustrate how family resources and natural talents affect the skill investments that generate observed earnings inequality. Finally, agency theory approaches earnings inequality from an entirely opposite direction. Instead of describing how worker decisions and endowments generate a distribution of individual productivities that gives rise to a distribution of earnings, agency models describe how firms choose a distribution of earnings in order to elicit desired levels of individual productivity. Highly skewed wage distributions within firms can be an incentive device.

After discussing these models, we offer an evaluation of their relative contributions to our understanding of the main stylized facts about earnings distributions. We also comment on a specific shortcoming in the existing theoretical literature on earnings distributions.

1. The characteristic skew of earnings distributions

In the physical and biological worlds, distributions tend to be symmetric and bell-shaped. But in the social world, size distributions tend to be asymmetric and right-skewed. Thus, we begin with stochastic process theories and other models that focus specifically

on generating earnings distributions with long right tails. We then turn to models that address skewness while generating richer behavioral implications.

From the central limit theorem, we know that normal distributions arise from sums of many independent and identically distributed components, irrespective of how the components themselves are distributed. To apply that argument to earnings and other long-tailed distributions in economics, it is necessary to hypothesize that a person's earning capacity is generated as a product of independent increments, as in

$$y = \prod \epsilon_i, \quad (1.1)$$

where the ϵ_i 's reflect the myriad factors that determine personal productivity (Roy, 1950). With enough independent ϵ -components, $\log(y)$ tends to the normal distribution whatever the distribution of the ϵ 's (Aitchison and Brown, 1957). The distribution of the natural units (the antilog) has a long right tail. However, observed earnings distributions tend to have tails that are thicker and longer than the lognormal. For instance, the Pareto distribution has "infinite" second, third, and higher order moments and fits the upper tail of earnings distributions quite well.

1.1. Stochastic process theories

The stochastic process approach focuses on the "long-tail problem" (see especially the accounts of Steindl, 1968; Lydall, 1968; Pen, 1971). It builds up a "generating process" for the overall distribution from more elementary micro components. However, the components themselves have no behavioral content. A lognormal distribution is generated by a random walk in which the percentage change in a person's earnings is distributed independently of earnings itself. Such a process results in Eq. (1.1), but since an individual's earnings level is the product of random variables realized up to that time, such a process implies ever growing variance in the overall distribution of log earnings. Some stabilizing force that offsets the tendency for increasing variance is necessary for the process to converge to a stationary steady-state distribution.

Various economists devised different auxiliary stabilizing hypotheses to solve this problem (Kalecki, 1945; Rutherford, 1955; Simon, 1955; Wold and Whittle, 1957). Champenowne (1953, 1973) established the basic method. He replaced the random walk with a Markov chain where workers face identical, fixed transition probabilities of moving between exogenously defined (log) earnings classes. Markov chains also have the variance-increasing property unless the transition probabilities are appropriately restricted. Champenowne's theory controls the variance of earnings over time by restricting the average transition to be downward, toward lower incomes. Intuitively, the idea of intergenerational turnover of workers justified this restriction (such turnover is explicit in Rutherford, 1955): higher earning older workers drop out of the process at retirement and are replaced, in a sense, with lower earning new entrants. It is slightly astonishing that a simple specific form of Champenowne's restriction implies Pareto's law as the stationary distribution.

Singh and Maddala (1976) pursued a related hazard function approach. If $F(x)$ is the cumulative probability distribution of random variable x and $f(x)$ is its density, $1 - F(x)$ is called the survival function, and $h(x) = f(x)/[1 - F(x)]$ is its hazard (failure) rate. A decreasing hazard seems necessary to produce the kind of probability masses observed in the upper tails of earnings distributions. For example, the normal has a hazard that is strictly increasing in x , but the hazard of the Pareto distribution is strictly decreasing (it is $h(x) = \alpha/x$). The hazard of the lognormal is not monotone. It first increases and then decreases in x . Singh and Maddala produce a new class of distributions by specifying that the proportional hazard is logistic, or alternatively that the hazard itself follows a sech^2 law. These new distributions closely fit the overall distribution of family incomes.

1.2. Correlated increments and common factors

Another way to account for the extra concentration of mass in the tail (relative to the lognormal) is to allow the ϵ_i components in Eq. (1.1) to be positively correlated. The law of large numbers no longer applies, but it is intuitively clear why a multiplicative structure in Eq. (1.1) and positive correlation leads to a long right tail in the distribution of y . Think of Eq. (1.1) as a “production function” for earnings capacity, with the ϵ_i ’s regarded as “inputs”. Then a person who has 100 $\lambda\%$ more of each component input earns $(1 + \lambda)^n$ times more money, where n is the number of different factors that determine individual productivity. The effect of λ is magnified many times over when n is large.

For instance, consider a statistical factor-analytic structure with a single factor λ , interpreted as generalized ability or “IQ”. Suppose personal endowments of the specific ϵ -inputs follow $\epsilon_i = k_i(1 + \lambda)$, where k_i are independently distributed. The common factor λ induces positive correlation between ϵ_i and ϵ_j . Then Eq. (1.1) implies that personal productivity is $y = K(1 + \lambda)^n$, with $K = \prod k_i$. Even if λ is symmetric and normally distributed, the distribution of $K(1 + \lambda)^n$ has a long right tail, and the highest earners account for a disproportionate share of total earnings.

An example brings the point close to home. This kind of model accounts for the highly skewed distribution of journal publications among members of an academic profession. Remarkably few scholars produce the lion’s share of academic publications: the median number of career publications in economics is one, while the mean is much larger (Lovell, 1973). The distribution of professional citations is even more concentrated (Leamer, 1981). Interpreting each ϵ_i as the probability of carrying through on one of the large number of steps needed to bring a research project to successful publication, scholars who are slightly better in each dimension are many times more likely to succeed overall.

When the ϵ_i ’s are interpreted as factors of production rather than as probabilities, it is natural to expect that some inputs are more important than others. A weighted specification is preferred, such as $y = \prod (\epsilon_i)^{\alpha_i}$, where the α_i ’s are fixed weights that affect marginal products. If $\sum \alpha_i > 1$, positive correlation among the ϵ_i ’s produces a long right tail in the y -distribution. A small common increment is magnified overall.

However, if $\sum \alpha_i \leq 1$ there is no magnification and skew cannot arise in this way. There is a sense in which a form of increasing returns to scale produces the characteristic skew of earnings distributions.

1.3. Scale-of-operations effects and superstars

Attempts to study the determinants of high earnings in terms of power and span-of-control (Simon, 1957; Lydall, 1959) are closely related to scale economies. What is at issue is a person's "scale of operations" (Mayer, 1960; Tuck, 1954; Reder, 1968). Most personal economic activity occurs on a small scale, but in some activities the size of a person's market can be noticeable, even relative to the economy as a whole. The term "scale of operations" comes from the tendency for extremely wealthy individuals, such as founders of large companies, to control vast amounts of capital and wealth. Scale economies account for this. For if person A can invest a dollar more efficiently than person B, person A can likely invest \$1 billion better than person B and will end up controlling resources on a much larger scale. A small extra talent for control will command a large rent in the market equilibrium.

The executive labor market is a leading practical example of this effect. Control aspects of firms and management decisions contain "local public goods" elements because they affect the ways in which all resources within the organization are directed. Top level managerial decisions, such as what products to produce and how to produce and market them, affect the productivity of all resources employed by the firm. Just as a good soldier is not effective if fighting the wrong war, a talented and highly motivated worker is not socially productive if the organization is producing a good no one wants to purchase. Better decisions of this kind are worth a great deal when applied to a large organization. A person with a small edge in talent can have a very large marginal product.

The relevant theory describes how managerial talents are allocated to control functions in a market economy (Tuck, 1954; Grubb, 1985; Rosen, 1982)—how the executive labor market allocates talent. In essence, workers and the owners of the capital bid for the privilege of being managed by talented people. Thus, rents from greater capacity to make superior control decisions get transferred to managers. In equilibrium, more talented managers control larger or more valuable organizations. This is the principle reason that top-level managers and executives in large companies earn such high salaries (Rosen, 1992) and are typically found in the top percentiles of the earnings distribution.

The scale of operations effect extends to a broader array of services that have local public goods qualities for other reasons. In many transactions, the marginal costs of serving another customer are small. Though different customers are served simultaneously, access is rationed by price. The services of intellectual property often have these properties, especially when rendered through mass media. Use of the service by one person does not greatly diminish availability to others. An author who must personally tell his story to each person individually has much smaller chances of earning a high income than one who can, in effect, clone himself to many customers at a time by

writing the story down once-and-for all and duplicating it, at negligible cost, to others as a book, cassette or compact disc. Similar considerations apply to entertainers, actors, television personalities and newscasters, musicians and composers, artists, intellectuals, software developers, and athletes (Rosen, 1981). Typically the chances of success in these endeavors are rather small, but the rewards of the successful can be enormous because relatively few people are needed to satisfy the entire market. The small chances of great success attract many entrants, and the “stars” who are successful *ex post* appear to dominate their fields. They earn enough to put them well into the upper tail of the earnings distribution (Frank and Cook (1995) present many examples).

However, the vast majority of workers in the top percentiles of the earnings distribution are not stars, superstars or chief executive officers because scale economies in the services they render mean that few people are required to produce them. There are simply not enough of them to make much of a dent in the upper tail, even in a large economy such as the US, where the incomes and activities of such people receive so much attention and publicity. Highly successful professional practitioners in law, medicine and other professions occupy the upper tail much more frequently. Here, technology sharply constrains one’s personal volume of transactions. Doctors can see a relatively small number of patients per day, and lawyers can manage only a few cases at a time. Instead, the scale-of-operations effect works through total value. The most capable lawyers tend to be assigned to the largest claims, where a slight edge in talent can have huge financial effects on the value of resources in dispute (Spurr, 1987). Large claims bid up the fees of the most successful lawyers. The wealthiest people in need of medical services outbid others for access to the most capable physicians.¹

Sattinger (1993) discusses the assignment of workers to machines or other indivisible units of capital. Jovanovic (1998) combines this assignment problem with a model of vintage capital. In his framework, lower costs of upgrading machines are associated with increased earnings inequality because skilled workers benefit most when new technologies are more readily available.

2. Selection theory

In 1951, Roy sketched the basic structure of selection, but economists did not widely recognize the importance of his work until the 1970’s. Tinbergen’s (1959) version is more elaborate (and more obscure),² so we follow Roy (1951) here. The microeconomic content of “the Roy model”, as it is sometimes called, is minimal. It is based on a choice problem of the simplest kind. Its analytic interest lies in its aggregate market implications. By describing how different individuals sort across job categories in the

¹ Alfred Marshall (1947) briefly introduced this point 100 years ago.

² Specifically, Tinbergen specifies that tastes for different kinds of work follow a specific form that has never been pursued in the literature. Tastes have been treated in the literature on equalizing differences, but not specifically in the occupational or job choice literature.

economy at large, the model helps us understand the nature, determinants, and distribution of economic rents in labor markets. It also illustrates how selection may generate asymmetric, long-tailed densities for earnings distributions even when the distribution of potential earnings in each job is symmetric.

2.1. The Roy model

Consider an economy with n different types of jobs. Earnings are worker specific because worker productivity varies from job to job, and some workers are better at some jobs than others. Assume that supply and demand are equal in all job markets and examine the assignment of workers to jobs in market equilibrium. Let y_{ij} be the earnings capacity of worker i on job j . It is what worker i could earn in job j . Let y_i be the earnings actually observed for worker i . Maintaining the economic hypothesis that self-interested workers choose jobs that maximize their earnings, observed earnings for worker i are

$$y_i = \max(y_{i1}, y_{i2}, \dots, y_{in}). \quad (2.1)$$

The model is completed by specifying a joint probability distribution $f(y_{i1}, y_{i2}, \dots, y_{in})$ for earnings prospects in the population at large. Roy (1951) assumed that $f(y)$ is jointly normal. Mandelbrot (1962) assumed that $f(\cdot)$ follows the Pareto–Levy form. The observed earnings distribution, $g(y)$, is the transformation of $f(y)$ implied by Eq. (2.1).

This simple model yields surprisingly complicated outcomes. The observed distribution $g(y)$ is a mixture of conditional distributions resulting from maximization. Note that if n is large, Eq. (2.1) suggests the extreme value distribution (the distribution of the first order-statistic) as a possible stationary distribution. The extreme value theorem, like the central limit theorem, proves that the first order-statistic of independent and identically distributed parents tends to a unique limiting distribution, independent of how the parents are distributed. The stationary distribution is a double exponential. It is skewed and leptokurtic, characteristic of the upper tail of earnings distributions.

Though this idea is insightful, the extreme-value theorem cannot be applied here because the component distributions of earning potential y_{ij} in each job are not identically distributed (they have different means and variances) and may not be independent. It proves necessary to work through all the conditioning information to derive the observed distribution $g(y)$. Mandelbrot (1962) shows that $g(y)$ follows the Pareto–Levy distribution if $f(\cdot)$ is Pareto–Levy. In fact, this is the only case for which $g(y)$ is in the same family as its parent $f(\cdot)$. Otherwise the conditional distributions which compose it are truncated, and $g(y)$ is a complicated mixture that is not easily described in closed form.

The model is best illustrated when there are two jobs ($n = 2$). Suppose $f(y_1, y_2)$ is normal (or lognormal), with means μ_1 and μ_2 , variances σ_1^2 and σ_2^2 and covariance σ_{12} .

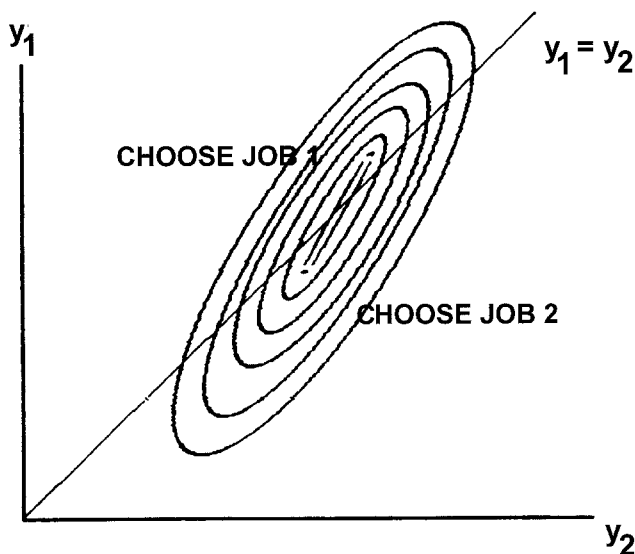


Fig. 2.1.

The correlation coefficient between earning capacities y_1 and y_2 is $\rho = \sigma_{12}/\sigma_1\sigma_2$. The assignment of workers to jobs is depicted in Fig. 2.1. The probability contours depicted assume $\mu_1 = \mu_2$, $\sigma_1 > \sigma_2$ and $\rho > 0$. Each person in the population is a point in the (y_1, y_2) plane. The probability mass there is proportional to the number of people in the population with those prospects. All workers along the 45° line ($y_1 = y_2$) find the two jobs equally attractive. Earnings in job 1 exceed those in job 2 for all people above the line. They choose job 1 and $y_i = y_{i1}$. Similarly, all people in the region below the 45° line maximize earnings by choosing job 2, so $y_i = y_{i2}$ for them. Therefore the observed distribution $g(y)$ is the weighted sum of two truncated normals, one above the 45° line and the other below it. Maddala (1977) and Heckman and Honore (1990) present complete details of this model.

Economic rents enter this model both in the sense of relative talents (“producer” surplus or comparative advantage) and in the sense of absolute talents (“ability” rents and absolute advantage). In Fig. 2.1, workers with prospects along the equal earnings line represent the extensive margin between job types. Relative to their alternative, they receive zero surplus from their actual job choice. All others are inframarginal, and workers furthest from the 45° line (in either direction) receive the largest surplus from their choices. The length of the (y_{i1}, y_{i2}) vector provides a measure of general ability. Holding the ratio (y_{i1}, y_{i2}) constant, earnings rise with general ability.

With a little stretch, the variances and covariance in the probability contours can be given an economic interpretation. For instance, $\sigma_1 > \sigma_2$ implies a sense in which job 1 is more difficult than job 2. In this case, the outcome in job 1 is more dependent

on who does the work. It offers more chances for talent to stand out. $\rho > 0$ means that a person whose earnings prospects are larger in job 1 is likely to have better than average prospects in job 2. For example, a one-factor structure of the type discussed above produces positive correlation between potential earnings in the two sectors and implies hierarchical selection. In Fig. 2.1 people with the highest earning prospects in both jobs tend to work in job 1. Those with lesser overall prospects tend to work in job 2. Notice that the relative variance condition is crucial to the order of the hierarchy. If $\sigma_1 < \sigma_2$ and $\rho > 0$, the probability contours cut the 45° line from above, and talented people are more often found in job 2.

The sign of the correlation coefficient is crucial to whether or not selection tends to be hierarchical. When $\rho < 0$, a person who is good at one job is likely to be below average in the other. The ellipses in Fig. 2.1 would be negatively sloped, and there would be positive selection in both jobs. Selection is not hierarchical in that case. Rather, people select jobs according to "comparative advantage". Selection on comparative advantage is empirically supported for the choice between high school and college education in some US data (Willis and Rosen, 1979). People choosing to stop school upon high school graduation are more likely to be attracted to blue collar jobs and those continuing through college are more likely to be found in white collar jobs. Willis and Rosen provide evidence that, on average, those who choose blue collar jobs are actually more productive in those jobs than more educated white collar workers would have been had they chosen blue collar work. This result is plausible to the extent that white and blue-collar jobs use much different kinds of talent and to the extent that different kinds of talent are negatively correlated. In such circumstances, the relevant concept of ability for earnings is clearly multidimensional. Single factor measures like IQ scores may be quite misleading.

Computational complexities of estimating selection models rise quickly in n , so these models have only been used for broad classification of job types. However, Garen (1984), who provides an approximation for selection bias in an educational choice model with a continuum of education levels, offers support for the multi-factor interpretation of selection according to comparative advantage. Heckman and Sedlacek (1985) expand the typical bivariate discrete econometric model to more alternatives, and recent applications of their methods reveal increasingly positive covariance in the earnings potential of individuals across occupations in the US data for the 1980's and 1990's (Gould, 1996), a period when earnings inequality in the US increased markedly. Perhaps selection on comparative advantage has become less important and selection according to absolute advantage more important over time. For instance, there is an alleged connection between increasing wage inequality and the information-processing content and computer intensity of work (Krueger, 1993).

Another possibility is that complementarities among qualities of workers have become more important for job assignments lately (Kremer, 1993: 1997). Worker complementarities are present when increasing stratification of worker types into more homogeneous groups raises their total productivity. Such an effect might arise from indivisi-

bilities of tasks at the work-site. Increasing use of testing and other merit based methods for screening applicants to schools appear much more important now than in the past. However, much research remains to be done in this under-developed area.

2.2. Selection on ability and stratification

Figure 2.1 reveals that workers with similar earnings prospects choose the same type of job. When the number of job types n is large, one can examine interesting details of the assignments by thinking of talents as worker attributes that determine specific job productivity. Assume

$$y_{ij} = w_j \pi_{ij}, \quad (2.2)$$

where π_{ij} is the volume of the i th worker's output in the j th job, and w_j is the price per unit of worker output in job j (think of w_j as a piece rate). The choice criteria remains as in Eq. (2.1), so all pair-wise comparisons of w_j/w_k with π_{ik}/π_{ij} reveal a person's comparative advantage (Sattinger, 1975, 1993; Rosen, 1978). Furthermore, write

$$\pi_{ij} = \alpha_{1j} C_{1i} + \alpha_{2j} C_{2i} + \cdots + \alpha_{Kj} C_{Ki}, \quad (2.3)$$

where the C_{ki} 's are individual worker characteristics such as mathematical, motor, and verbal skills, etc., and the α_{kj} 's are the marginal products of those traits for the specific job. These marginal products generally vary from job to job. For instance, verbal skills are of greater value to a lawyer than to a carpenter, but manual dexterity is of greater value to a carpenter than to a lawyer. Combining Eqs (2.2) and (2.3),

$$y_{ij} = \beta_{1j} C_{1i} + \beta_{2j} C_{2i} + \cdots + \beta_{Kj} C_{Ki}, \quad (2.4)$$

where $\beta_{kj} = w_j \alpha_{kj}$ is the value of the marginal product of characteristic k in job j . Idiosyncratic random terms could be added to Eqs (2.2)–(2.3), but we ignore them here.

Applying the choice criterion in Eq. (2.1) to Eq. (2.4) reveals how people with similar C -endowments tend to choose the same type of job. Figure 2.2 illustrates the equilibrium stratification for 5 job types and two characteristics ($n = 5$, $K = 2$). Given job j , all people whose characteristics lie along the downward sloping line $C_1 = Y^*/\beta_{1j} - (\beta_{2j}/\beta_{1j})C_2$ would earn the same amount of money Y^* if they chose to work on job j . Draw such a line in the (C_1, C_2) plane for every job type and consider the inner envelope of the entire family of lines. This is the negatively inclined piecewise linear curve in the figure, after ordering job numbers by β_{2j}/β_{1j} . Rays from the origin through the corners of the envelope define regions where workers endowed with such traits maximize their incomes by choosing jobs 1, 2, ..., or 5. These partitions are completely deterministic when there is no randomness in Eq. (2.3). If specific random errors (noises) are present in Eq. (2.3), the partitions in Fig. 2.2 represent assignments in expected value. The partitions of the space of characteristics in Fig. 2.2 are open cones

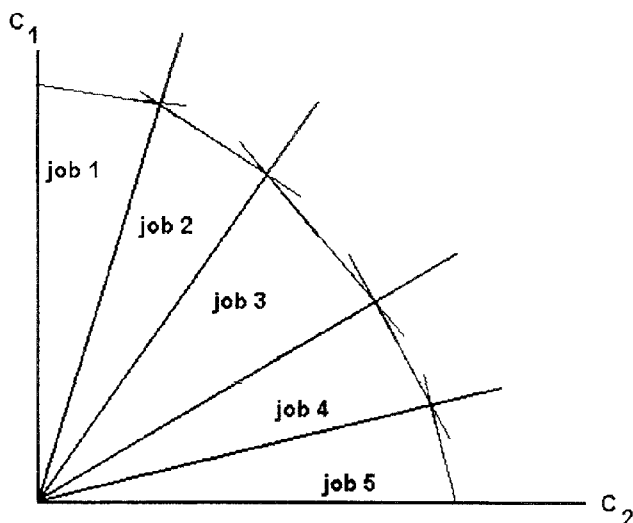


Fig. 2.2.

when the relations in Eq. (2.3) are homogeneous. However, if Eq. (2.3) has constant terms, the partitions can be closed and are much more complicated to describe.

Workers tend to choose jobs that highly value the personal characteristics they possess in relative abundance. The slope of each linear segment of the envelope in Fig. 2.2 is the ratio of marginal products α_{2j}/α_{1j} in Eq. (2.3). As illustrated, characteristic C_1 is highly productive in job 1 and C_2 is highly productive in job 5. Workers with relatively greater endowments of C_1 tend to choose job 1 and those with relatively greater endowments of C_2 tend to choose job 5. For instance, if C_1 is verbal skill and C_2 is manual dexterity, job 1 might be lawyer and job 5 might be carpenter. Both characteristics are equally important for productivity in job 3 and it tends to attract people with more balanced endowments. In this model, large earnings require large personal endowments of the C 's. To the extent it is less probable to have large endowments of many characteristics than of only a few, high earners will tend to be more frequently found in occupations such as 1 and 5, which weigh only one characteristic strongly (Mandelbrot, 1962).

Prices, w_j , also affect the size of the partitions in Fig. 2.2. For instance, an increase in w_j reduces the intercept of the line that defines the facet and expands the width of the cone for which occupation j is optimal. Some persons who otherwise would have chosen neighboring occupations $j \pm 1$ now choose occupation j . The supply price of workers to jobs is always rising.

2.3. Factor price equalization?

The selection model contrasts with the Lancasterian combinable characteristics (hedonic) approach to product differentiation. Both place economic structure on a statistical factor-analytic model of personal earnings. In the hedonic approach, each worker supplies certain factors of production to employers, like strength and intelligence, etc. Equating demand and supply of each factor in an implicit aggregate factor market establishes market equilibrium prices and fixes the factor-loadings on the statistical structure. A worker's labor income is a fixed weighted sum of personal factor endowments (plus noise). If the number of factors is relatively small, the earnings distribution is compacted into a space of equally small dimensions. In effect, there are as many basic types of workers as underlying factors, and the earnings of every worker can be expressed as a linear combination of basic types (Welch, 1969)

This method rests on the crucial assumption that the total amounts of worker characteristics employed by a firm or sector of the economy are combinable and serve as its factors of production, independent of how the totals are obtained. It is a multivariate generalization of the old "efficiency units" assumption; e.g., a firm hiring two workers with half as many characteristics is the same as hiring one worker with twice as many. In essence, workers can unbundle their skills. Workers are allowed to allocate one type of skill to a given task while simultaneously allocating other skills to different tasks. In this form, the model is isomorphic to the theory of international trade, and the main question for its empirical relevance is whether or not there is "factor price equalization" across firms and industries. Unique prices imply that factor loadings are the same for all workers. But, if factor price equalization fails, the factor loadings are job specific, and the dimensionality economies afforded by the hedonic approach are lost.

Taking this model on its own terms, factor price equalization in labor markets is improbable. It fails if specialized factors are used in some firms but not in others. Further, because workers' skills are embodied, they cannot sell their skills in separate markets. An agent cannot sell brains to one employer while selling braun to another. This restriction limits the scope of arbitrage activities required to ensure unique factor prices in the labor market. The failure of factor price equalization makes the model in Fig. 2.2 more relevant (Rosen, 1983a; Heckman and Scheinkman, 1987).

3. Learning, sorting and matching

In the models discussed to this point, agents know their own productive capacities. We now review models where both workers and firms are uncertain about worker endowments. Both learn about the expected productivity of particular workers through successive observations of their performance. These models are dynamic extensions of selection and have implications for how the size distribution of earnings within a cohort evolves over time.

In simplest learning models, agents do not use information from the record of previous output when making subsequent allocation decisions. Information only influences wage rates. Since each observation of a person's output is an error-ridden measure of productivity, labor market competition implies that workers are paid their expected marginal products, given the information available. Thus wages for a given worker evolve as information accumulates over time. If the record is long enough, earnings in a cohort converge to the stationary distributions described in the stochastic models of Section 1.

More economically interesting information models introduce feedbacks from the data to future allocations along the lines of the selection models of Section 2. In sorting models, workers learn about their comparative advantage in various jobs and can switch jobs over time to better exploit their most valuable traits. In matching models, workers and firms gradually learn how productively they match with each other. If a realization is unfavorable, the relationship is terminated, and each side seeks a new match. Because workers keep good matches but abandon bad matches, wages grow, on average, as workers gain more information through work experience. Further, because workers leave poor matches, the wage distribution for a cohort becomes truncated over time, and wage distributions for experienced workers tend to be skewed relative to the distribution of wages among new workers.

Both matching and sorting models capture important aspects of individual careers. Without information about their relative aptitudes for different types of work, workers cannot make occupation choices that maximize their earnings. Thus, workers receive two forms of compensation from work experience. They receive their current earnings, and they receive information about their skills. This information affects future earnings because it informs future job choices.

Analyzing dynamic "Roy Models" requires putting more detailed structures on individual choices, and these structures make it more difficult to study the market equilibrium aspects of the problem. Little work has been done on how individual decisions based on gaining better personal information link up to overall market behavior. For instance, most learning models implicitly assume infinitely elastic demands for all types of skills. An exception is MacDonald (1982), who nests a learning model inside a supply and demand framework. MacDonald illustrates how learning generates sorting of workers across tasks in a general equilibrium context. The sorting process in his model also generates aggregate wage growth for a cohort and skewness in the size distribution of wages.

3.1. *The basic learning model*

Assume that two individual traits, X_i and θ_i , affect personal productivity of individual i at time t , y_{it} . All agents observe X_i . No one observes θ_i directly (workers have no private information). In each period, nature randomly assigns workers to firms. They are paid their expected marginal product as assessed at the beginning of the period, given X_i and their productivity history, $y_{it-1}, y_{it-2}, \dots, y_{i0}$. The realization of worker output in each

period is subject to an i.i.d. shock. Thus, output is a noisy measure from which to infer θ_i . Repeated observations of past performance allow inferences to become sharper over time. At each point, the worker's output y_{it} is an independent draw from the distribution $G(y_{it}|X_i, \theta_i)$.

Farber and Gibbons (1996) study how the distribution of wages evolves for a given cohort of workers in these circumstances. This problem is greatly simplified by the fact that true productivity does not vary over time. Consider a group of workers who share the same value of X_i . Since y_{it} is a random sample out of a fixed distribution, the mean wage for this group of workers does not change as the workers age. There is no cohort wage growth in large samples, and the mean wage in any period among these workers is always

$$E(w_{it}|X_i) = E(y_{it}|X_i) = E[(y_{it}|X_i, y_{it-1}, y_{it-2}, \dots, y_{i0})|X_i]. \quad (3.1)$$

However, from well-known results on signal extraction in statistical decision theory, the variance of wages within each X_i class, and hence overall, increases with cohort experience. Simple as it is, this model produces the fundamental and important result that the distribution of wages is more compressed among young cohorts because there is not much information to differentiate one from another. Wage dispersion increases as work records reveal each worker's productivity more precisely.

Learning models are more interesting (and more difficult) when agents use acquired information to make real allocative decisions. The literature treats two kinds of problems. In one, workers use information to decide to invest in specific skills depending on what they learn about themselves (Jovanovic, 1979b). The implications for earnings distributions are perhaps not distinctive enough from the generic human capital models considered in Section 4 to warrant discussion here. In the other, matching and sorting models, workers base job mobility decisions on information gained from previous work experience. This kind of learning does not change skills per se. Rather, it changes perceptions of what skills one has and how they may be used more effectively. While these models occupy an important place in the literature on job mobility, their implications for individual wage growth also yield important insights concerning the evolution of the distribution of earnings for a particular cohort of workers.

A basic economic idea in information theory is that agents anticipate a priori how information gained currently and in the future will affect their subsequent decisions. For instance taking a low paying, low productivity job might have value in providing much information about the best sequence of subsequent jobs. The ability to change one's decision as information accumulates lends option value to choices. But this is very difficult to characterize analytically, and the structure of most work in labor economics minimizes this aspect of the problem.³

³ Miller (1984) is an exception. Miller shows how workers sort across occupations when jobs differ with respect to both current expected returns and information content. In Miller's model, experience in a given job provides no information about expected productivity in other jobs, and given this form of independence across

3.2. Sorting by comparative advantage

Consider the following problem. There are N workers and J possible jobs. In each period output, y_{ijt} , of worker i on job j is

$$y_{ijt} = h(\theta_i, X_i, Z_j, \epsilon_{ijt}), \quad (3.2)$$

where θ_i are unobserved traits, and X_i are observed traits as before, Z_j is a set of job attributes, and ϵ_{ijt} is an idiosyncratic i.i.d. shock drawn from a known distribution $G(\epsilon)$. The function $h(\cdot)$ is known, and outputs, y_{ijt} , are observed publicly. The “person effect” θ_i is an i.i.d. draw from a known distribution $F(\theta)$. Wage setting goes as before, and workers and firms are risk neutral. In the case where job characteristics are known, wages are given by

$$w_{ijt} = E(y_{ijt} | X_i, Z_j, y_{ijt-1}, \dots, y_{ij1}). \quad (3.3)$$

Workers choose the sequence of jobs that maximizes the expected present value of lifetime earnings, $\sum \beta^t w_{ijt}$, where β is the discount factor.

The solution is trivial when all traits of jobs and individuals are known: pick the job with the highest expected output and never switch. However, if some traits of jobs or individuals are unknown, then workers may sort across jobs based on their observed record of productivity in various employments. Sorting models describe how workers learn about their unobserved traits, θ_i , when these traits have different values on different jobs. Information helps allocate workers to these jobs more efficiently over time. By contrast, matching models describe situations where all learning is match-specific. Work experience on a particular job provides no information about the distributions of potential rewards associated with other jobs.

Matching models are easier to analyze, but sorting models may be more realistic. Nonetheless, both approaches tell the same basic story. As workers learn about their comparative advantages in different types of employment, or about the quality of their current job match, they sort away from things that they do poorly. This implies that, for a given cohort, average wages grow with experience. Further, in the sorting model below, the truncation of distributions associated with selection generates skewness in the distribution of wages.

The sorting model of Gibbons et al. (1997) nicely illustrates the main ideas while finessing many technical dynamic programming difficulties.⁴ Here, the function $h(\cdot)$ in Eq. (3.2) has the form

$$y_{ijt} = d_j + c_j(\theta_i + \epsilon_{ijt}). \quad (3.4)$$

jobs, he is able to use results from the literature on Dynamic Allocation Indices (or Gittins indices) to describe a tractable solution to the model. In models without this type of independence, tractable solutions are unlikely unless the information content of all jobs is the same.

⁴ See Gibbons and Waldman (1998) for a related model dealing with promotion decisions. Kim (1998) and Ross et al. (1981) also provide variants of this type of model.

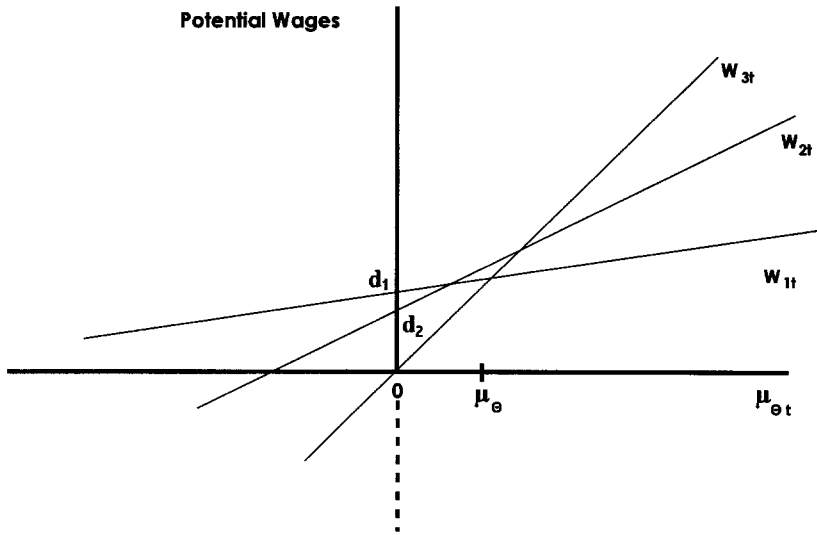


Fig. 3.1.

As in Section 2, this is a one-factor structure on “worker ability” θ_i , with factor loadings differing across job types. Here, the Z_j ’s in Eq. (3.2) are two parameters, c_j and d_j , and a known pair $\{c_j, d_j\}$ characterizes each job. Workers have one trait, θ_i . We ignore observed individual characteristics, X_i .

Coexistence of all types of jobs requires that if $c_j > c_{j'}$, then $d_j > d_{j'}$. Figure 3.1 illustrates three jobs. Job 1 requires almost no skill. It pays wages that vary little with worker ability. Job 2 offers higher returns to ability, and job 3 offers the highest returns to ability. The shocks, ϵ_{ijt} , come from a standard normal distribution. Individual ability, θ_i , also comes from a normal distribution with mean μ_θ and standard deviation σ_θ . Notice from the figure that if agents know θ_i , the analysis reduces to the deterministic factor model in Section 2, and sorting is strictly hierarchical: people with the highest values of θ_i choose job type 3 and those with the lowest values choose job type 1.

Hierarchical sorting simplifies the problem.⁵ The problem is further simplified because the specified technology ensures that the information content of all jobs is the same and therefore does not affect job choice. Agents base posterior beliefs about the distribution of a worker’s value of θ_i on the statistic

$$V_{ijt} = \theta_i + \epsilon_{ijt} = (y_{ijt} - d_j)/c_j. \quad (3.5)$$

⁵ As noted in Section 2, if workers differ with respect to more than one skill, the presence of constant terms in Eq. (3.4) complicates sorting partitions.

Since the distribution of V_{ijt} is invariant across jobs j , the worker's optimal policy is to pick the job that offers the highest wage in each period. This is given by

$$\max_j w_{ijt} = C_j \mu_{\theta_{it}} + d_j, \quad (3.6)$$

where $\mu_{\theta_{it}}$ is the posterior mean of θ_i based on information available up to time t .

Consider a single cohort of workers. Assume these workers initially have the same expected ability, in the sense that their θ 's come from the same distribution. The optimal policy described in Eq. (3.6) generates three implications for the evolution of the distribution of wages for this cohort. First, the distribution of wages becomes more dispersed as the cohort gains experience. Second, as workers sort across jobs in response to the information they receive, the average wage for the cohort as a whole increases with experience. Third, sorting across jobs generates skewness in the distribution of wages.

To understand how the sorting of workers across jobs creates wage growth in this model, note that since both θ_i and ϵ_{ijt} are normally distributed, $\mu_{\theta_{it}}$ is normally distributed across workers for $t > 1$. Further, the mean and median of $\mu_{\theta_{it}}$ equals μ_θ in each period. Consider the job choice of a worker with $\mu_{\theta_{it}} = \mu_\theta$ in Fig. 3.1. This worker chooses $j = 2$. If this were the only job in the economy, wages would be determined by

$$w_{it} = c_2 \mu_{\theta_{it}} + d_2, \quad (3.7)$$

and the variance of wages would increase with t , but the mean would remain unchanged at the starting wage, $w_{\text{start}} = c_2 \mu_\theta + d_2$. This is just a version of the pure learning model discussed in the previous subsection.

With more than one job, workers still begin their careers in $j = 2$, and all receive the same first period wage. But now, $\mu_{\theta_{it}}$ is not the only wage determinant that changes over time because workers have the opportunity to change jobs. Those who get favorable information and remain on $j = 2$ see their wage go up. Those who get unfavorable information and remain on $j = 2$ see their wage fall. However, the ability to change jobs convexifies the mapping from ability to wages, because those who learn they are much more capable move to $j = 3$, while those learning that their value of θ_i is quite small switch to $j = 1$, where ability is not so important. Those who switch from $j = 2$ to either $j = 1$ or $j = 3$ earn strictly more than they would have earned if they stuck with $j = 2$. Thus, the expected wage for a cohort of experienced workers exceeds the starting wage. Convexity of the market payoff to ability also imparts skew to the wage distribution. In sufficiently large samples, the median wage is constant. Therefore, the mean wage growth that results from sorting creates a growing gap between the mean and median wage.

The same logic applies in a model with an arbitrary number of jobs. As long as some workers sort away from their starting job, wages will grow on average with worker experience, and the distribution of wages will be skewed. In addition, since the variance of expected ability $\mu_{\theta_{it}}$ increases monotonically with t , the fraction of workers who have

switched to new jobs increases monotonically with t . This implies that the gap between the mean and median wage of a cohort increases with experience. The truncation of distributions with sorting makes it difficult to say how the variance of wages changes with cohort experience. However, it can be shown that interdecile range measures of dispersion increase with t .⁶

To put models of this kind into perspective, bear in mind that much learning about individual talents (and preferences) occurs in school, prior to labor market entry. This is one of the reasons why all workers do not start work at the same wage on the same job. Instead, workers begin their careers on a range of jobs at a range of wages. The point here is that both of these ranges expand as workers gain experience. This model provides a useful description of how work experience yields information that affects the sorting of workers across jobs and how sorting affects the evolution of the distribution of wages for a given cohort of workers.

3.3. Matching models

In matching models, all learning is idiosyncratic to a particular worker-job match. Experience in one job provides no information about a worker's potential productivity in any other jobs. The following sketch draws heavily on Jovanovic's (1979a) model, which in turn resembles a search model. Worker i 's productivity in job j at time t equals

$$y_{ijt} = \eta_{ij} + \epsilon_{ijt}. \quad (3.8)$$

Agents observe y_{ijt} , but do not observe its components. η_{ij} is the expected value of the match between worker i and job j , and ϵ_{ijt} is an i.i.d. realization of a normally distributed shock. The matches, η_{ij} , are i.i.d. draws from a normal distribution with mean μ_η and standard deviation σ_η . They are specific to workers and jobs jointly, not to either one separately as in sorting models. Again, workers are paid at the beginning of each period, before they produce output, and agents are risk neutral. There are enough firms competing for workers in all job categories that the labor market is competitive: the wage for worker i in job j equals the expected value of the match, η_{ij} , given past outputs.

As in search theory, the worker's optimum policy is to choose a reservation wage and switch jobs if the expected value of the current match, given all available information, falls below it. The reservation wage is a function of μ_η , σ_η , and tenure in the current job, τ . A job change in this model occurs when an agent draws a new η_{ij} . Average wages for a cohort grow with experience because workers and firms have the option to abandon bad matches. Further, the reservation wage policy ensures that, in the limit, the distribution

⁶ For example, the 90–10 ratio increase with t because the variance of $\mu_{\theta_{it}}$ increases. As workers gain experience, both the 90th and the 10th percentiles of the $\mu_{\theta_{it}}$ distribution are moving farther away from the median μ_θ , which is constant over time. Since wages are a strictly monotonic function of $\mu_{\theta_{it}}$, the 90th and 10th percentiles of the wage distribution must also be moving farther away from the mean wage, w_{start} .

of wages is truncated. The left tail of the distribution is cut off at the reservation wage for a worker who knows η_{ij} .

3.4. Learning and general equilibrium

Most sorting and matching theory is partial equilibrium and does not analyze how the number of people engaged in an activity affect its returns. MacDonald (1982) embeds a sorting model in a market equilibrium context, and his results reinforce the predictions of basic sorting models. Wages grow, on average, with worker experience, and the sorting process tends to generate skewness in the distribution of wages.

MacDonald adapts the comparative advantage structure of Sattinger (1975) and Rosen (1978) to a learning environment in which workers learn their abilities over time. The structure of production is such that individual workers perform a large number of tasks arranged along a continuum, indexed by s ; $0 \leq s \leq 1$. There are two types of workers, A and B, but neither firms nor workers know the type of a specific person. Workers of types A and B differ in their ability to perform the tasks. Type A workers have comparative advantage at one end of the spectrum, and type B workers have comparative advantage at the other end.

A fraction δ of all workers are actually type A workers. At the end of a period, each worker exogenously receives a piece of public information; an independent realization of a random variable X , where X takes one of two values. $X = a$, or $X = b$. The probability that $X = a$ is $\theta \geq \frac{1}{2}$ if the worker is an A. Likewise, the probability that $X = b$ is also θ if the worker is a B. All workers work for $N + 1$ periods and then stop. Thus, given the information technology, every worker belongs to one of $2N + 1$ information classes, where an information class is a group of workers with the same posterior probability of being of type A.⁷ At any point in time, the supply of workers in a given information class is determined by the age distribution in the population.

MacDonald illustrates the determination of firm demands for workers from each information class in two steps. Given any set of workers from various information classes, the firm must determine the best way to assign workers to tasks. The solution to the assignment problem involves a set of critical tasks that define intervals on the task continuum, with one interval per information class. Because comparative advantage is ordered when there are only two types, the critical task intervals are ordered. Information classes are matched with these intervals according to the probabilities that workers in the class are of type A. These information classes correspond to conventional factors of production. Given an array of market wages for workers of various information classes, the firm must also choose the combination of optimally assigned workers that minimizes the cost of producing output. The solution yields the cost function and the

⁷ Consider workers who vary with respect to experience, n . Denote x_n as the number of times $X = a$ in a work history with n periods. MacDonald shows that the number $k = n - 2x_n$ defines an information class. Workers who vary with respect to experience and x_n belong to the same information class if they share the same value of k . Figure 1 in MacDonald's paper shows that k may take on $2N + 1$ distinct values.

associated compensated factor demands. Markets clear when the demand for workers in each information class equals their availability in the population. Overlapping cohorts lead to a steady state equilibrium.

Equilibrium wages are positively correlated with worker experience in this model. As the market acquires more information about a given cohort of workers, these workers are, on average, assigned to tasks that more fully exploit their true comparative advantage, like the sorting model described above. However, the result is reinforced in this model by supply conditions. In this economy, young workers are found in information classes that contain relatively large stocks of workers, both within each cohort and in the labor market as a whole. Thus young workers have many substitutes. This is a second reason that young workers, on average, earn low wages.

For most interesting parameter specifications, the size distribution of wages in this model is skewed. If the skill endowments are such that neither A nor B has an absolute advantage at all tasks, the distribution of wages is always skewed. Further, in the benchmark case $\delta = 1/2$, mean wages exceed the median wage, regardless of initial skill differences between types A and B. Here, the median is constant, and mean wages grow with cohort age as workers are matched with tasks that better exploit their comparative advantage. However, if type A workers are more productive in all tasks, and $\delta \neq \frac{1}{2}$, then the distribution of wages may or may not be skewed. For example, if A workers are more productive in all tasks, and $\delta > \frac{1}{2}$, then median wages will grow with cohort experience because more than half of the workers in a cohort are actually “good” workers. In such a case, it is difficult to make general statements about the relative growth of cohort means and medians.

3.5. Summary

We have reviewed three types of models that highlight the role of information in determining wages. Simple learning models illustrate how wages should become more dispersed for a given cohort as the market gathers more information about cohort members. Sorting and matching models are an important part of the literature of wage growth and labor mobility, but the wage growth mechanisms described in these models also have implications for the evolution of the distribution of wages for a cohort. Sorting models provide the most direct contact with static theories of the distribution of earnings because they add a dynamic dimension to the literature on selection models. In both sorting and selection models, wage distributions are skewed relative to distributions of wage offers because workers sort to jobs where they have a comparative advantage. However, in sorting models, workers learn their comparative advantage only with time and experience. This implies aggregate wage growth for a cohort, and often implies that the distribution of wages for a cohort becomes more skewed as the cohort ages.

4. Human capital

In *The Wealth of Nations*, Adam Smith stated the central idea in modern human capital theory,

When any expensive piece of machine is erected, the extraordinary work to be performed by it before it is worn out, it must be expected, will replace the capital laid out upon it, with at least the ordinary rate of profits. A man educated at the expense of much labor and time . . . may be compared to one of those expensive machines. The work which he learns to perform, it must be expected, over and above the wages of common labour, will replace to him the whole expense of his education with at least the ordinary rate of profits of an equally valuable capital.

Some wage inequality is a consequence of the fact that employments differ greatly with respect to the “difficulty and expense of learning the business”. Because wages must compensate workers for the cost of learning their trade, wages would differ across employments even in a world where all workers possess the same initial skill endowments. The monetary value of the effort and time required to learn neurosurgery is great. If the wages of brain surgeons did not greatly exceed the wages of workers who learn their craft easily and quickly, we would observe a dire shortage of brain surgeons.

The two previous sections highlight how the market assigns people with heterogeneous endowments to heterogeneous tasks and how assignments change as markets reveal information about workers’ true endowments. Many models of human capital investment also highlight the role of heterogeneous endowments, but the most important insight of the human capital approach is that patterns of investment in training help determine differences in productive capacities across workers. Here, investment costs drive earnings differences across groups of workers who have different levels of education, experience, or professional training.

4.1. Compensation for investments in training

Mincer (1958) is among the first modern economists to formalize Adam Smith’s ideas concerning wage inequality and occupational training requirements. Suppose a worker who seeks to maximize his lifetime earnings is faced with two possible careers. The first requires d periods of training and pays a wage of W_0 per period. The second requires training for $d + s$ periods and pays a skilled wage of W_1 per period. Training requires no direct expense. The worker lives for n periods and discounts the future at rate r . A worker is indifferent between the two occupations if they offer the same lifetime earnings, or

$$\int_d^n W_0 e^{-rt} dt = \int_{d+s}^n W_1 e^{-rt} dt. \quad (4.1)$$

The ratio W_1 / W_0 that leaves the worker indifferent between the two occupations is

$$k(d, s, n) = \frac{W_1}{W_0} = \frac{e^{-rn} - e^{-rd}}{e^{-rn} - e^{-r(d+s)}}, \quad (4.2)$$

k is the relative supply price of labor to the schooling-intensive occupation. It is increasing in s and decreasing in n . Though schooling has no direct cost, an increase in s postpones entry into the labor market and increases foregone earnings: k must increase for the worker to remain indifferent. Further, since an increase in work life, n , implies a longer working period in which to recover foregone earnings, k decreases.⁸ Finally, k is an increasing function of d . Relative wage ratios and the rate of return to additional schooling should be larger for persons with 14 years of schooling than for those with 10 years, because when life is finite and independent of investment, a higher relative wage premium is required to cover the costs of schooling that occurs later in life. These last two effects are small when n is large.

As n goes to infinity, $k(d, s, n)$ ceases to be a function of d or n . In the limit (Mincer, 1974)

$$\ln W_1 = \ln W_0 + rs. \quad (4.3)$$

This is the clearest statement of how wages compensate for added training costs. Let s be the additional schooling required to perform job 1 instead of job 0. A worker is indifferent between the two jobs if the percentage difference between the wage for job 1 and the wage for job 0 is roughly equal to the product of r and s . If a group of workers, who are homogeneous with respect to talent and tastes, occupy a variety of occupations that require different levels of training or schooling, the distribution of wages for trained workers in this group is skewed relative to the distribution of schooling. For example, if the distribution of schooling is normal, the distribution of wages is lognormal.

Education and training means that wage inequality is not exclusively driven by heterogeneous endowments and rents. Wage inequality results when some occupations require more training than others. Wage differences also must equalize the net advantages of work in other ways (see the survey in Rosen, 1986). For instance, some jobs are more onerous or dangerous than others. Some offer more earnings stability than others. Though these important factors affect measured inequality indexes, the scope of the subject is too broad to be included here. However, education and training deserve special emphasis for this survey because they are so important in determining personal productivity.

Mincer's approach is valuable because it illustrates an equilibrium condition that must be satisfied when workers acquire skills by investment. But, it does not tell us which workers invest in training and under what circumstances. We next review models of optimal investment in human capital to shed more light on the mapping between the distribution of individual traits and the distributions of wages and earnings.

So far, distinguishing between wage rates and total earnings has been inconsequential. In models discussed to this point, workers decide where to allocate their time, but,

⁸ In societies where people work until roughly the end of their lives, increasing expected life span is associated with decreases in the relative supply price to training-intensive occupations. This effect has been important historically.

by assumption, workers allocate all of their time to some type of production. Further, in Mincer's original paper, while there is a training period, it is distinct from work, and once training is over, workers spend all their time producing goods. By contrast, in most human capital models, workers decide how to blend different activities at each point in time. Some models address the division of time between training and production. Others also address the consumption of leisure. In all cases, earnings, which equal incomes per unit of time, need not equal wages, which equal incomes per unit of time spent in goods production.

Having said this, we discuss the implications of human capital models in terms of actual earnings not wage rates or potential earnings. Links between the predictions of human capital models and empirical regularities in labor market data are more clear in earnings data than in wage data. Existing data often provide straightforward measures of actual earnings, but this is not always the case with hourly wage rates.⁹

4.2. *Optimal investment*

The most well known attempt to formulate a model that captures the implications of optimal investment in human capital for the distribution of labor earnings is Becker (1967),¹⁰ who developed the useful graphical representation in Fig. 4.1. In Becker's model, workers supply labor inelastically and seek to maximize the present value of their lifetime earnings. Great simplification is achieved by reducing lifetime human capital investment to a static problem. Assume that human capital investment activities require complementary purchased inputs (e.g., books and teachers) as well as the student's time and that each investment activity combines time and purchased inputs in fixed proportions. Workers acquire human capital by progressing through a sequence of investments. Returns on successive investments must diminish at some point because the remaining work life is shortened and because the worker's opportunity cost of time increases with previous investments in human capital.¹¹

The horizontal axis in Fig. 4.1 measures total lifetime expenditure on inputs purchased for human capital production. The downward sloping "demand" curve represents the marginal internal return on these expenditures. It equals $(PV_1 - PV_0 - C_1)/C_1$, where PV_1 and PV_0 are the present values of lifetime earnings with and without the marginal investment in human capital, and C_1 is the dollar value of purchased inputs

⁹ For example, consider a college student who works for only three months in the summer and reports annual earnings of \$5,000. Researchers commonly conclude that the student's summer earnings imply an annual wage rate or potential annual earnings of \$20,000. Among other things, this conclusion rests on the assumption that the student receives no training on the summer job. If the student is working in an apprentice program or an internship, his actual earnings may grossly understate his earning potential which is based on the wage he could receive in a job that involved no training.

¹⁰ This was later reprinted in Becker (1975).

¹¹ There may be regions of increasing returns if capital accumulation facilitates future learning. Below, we discuss assumptions concerning the way in which existing human capital affects the creation of new human capital. These assumptions are key to pinning down exactly how human capital theory links the distribution of talent and the distribution of earnings.

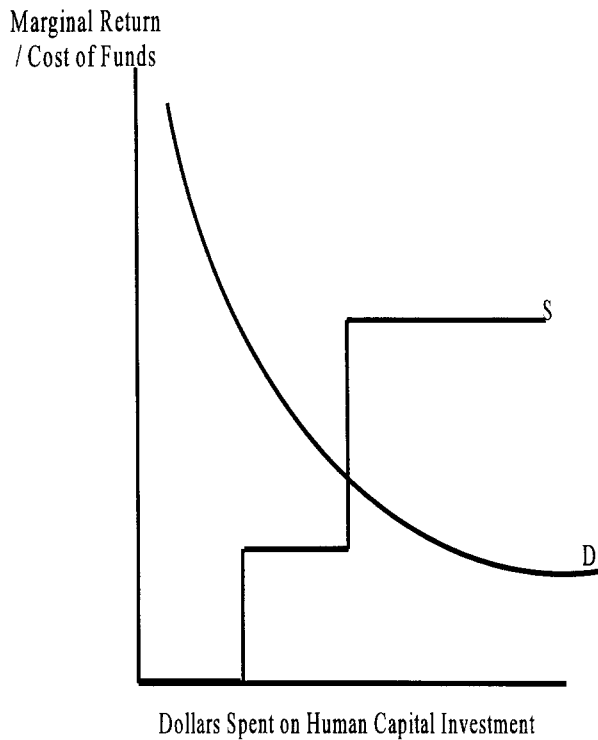


Fig. 4.1.

for the marginal investment.¹² Note that because $(PV_1 - PV_0)$ is the difference in the lifetime present values of two earnings streams, the marginal return in Fig. 4.1 is a net rate of return on the financial cost of investment, C_1 . This rate of return incorporates the opportunity cost of foregone earnings.

The upward sloping “supply” curve traces out the opportunity cost of purchased inputs in human capital production. In most industrialized countries, elementary and secondary schooling are approximately free in terms of tuition costs, but investments in on-the-job training or post-secondary schooling require other inputs that families must pay for with their resources. For wealthy agents, the opportunity cost of funds spent on these purchased inputs is the market rate of interest. For poor agents, the price of these funds is the interest rate at which they can borrow to finance their human capital investments. The supply curve of funds for poorer agents is generally upward sloping but may have horizontal segments. Optimal lifetime investment is determined by the intersection of the supply and demand schedules.

¹² These present values are calculated for all individuals and investments using a common market interest rate.

Figure 4.1 pushes a number of important details into the background. Nonetheless, it is a useful general description of how ability and family wealth interact to determine the distribution of lifetime earnings. In a rough and ready way, the “demand” and “supply” representations in the figure correspond to the concepts of “nature and nurture” that dominate popular debates on inequality. The relevant concept of ability here is associated with returns from human capital investments. Assume that two workers, A and B, consider the same sequence of potential investments. Worker A is more able than worker B if A’s marginal return from any particular human capital investment is always greater than B’s. The demand curve for human capital by worker A is shifted up relative to worker B, and given the same supply conditions, worker A invests more than worker B. The effects of family circumstances and credit market imperfections work through the supply curve. Students with wealthier parents can finance their investments on better terms. Their supply curves are farther to the right than those of students from poorer backgrounds. Holding ability constant, students from wealthier families invest more.

Superimposing the demand and supply functions of all individuals on the same graph yields a cloud of points that describes the entire distribution of life-investment outcomes. For instance, the distribution of lifetime earnings tends to be more skewed when people have different talents (demand curves) and the same opportunities for financing investments (identical supply curves) than when people have identical talents but differ in their access to funds. Becker calls the first case “equal opportunity”, because the terms of investment are independent of personal financial circumstances and family wealth. Those with large total investments in human capital also have high returns at the margin. But if all agents have the same ability and face different costs of funds, family wealth dominates investment outcomes: those with large total investments have low marginal returns because their marginal costs are low. Policies, like free education, that promote equal opportunity reduce the dispersion in supply curves.

This model has limited empirical content for understanding the details of observed earnings distributions because it deals with unobservable human capital concepts and does not impose enough structure to pin down the precise mapping from human capital wealth to observed earnings. This difficulty is transparent when one compares the model to the pure equalizing difference model of Mincer (1958). In Mincer’s model, all persons have the same talent and tastes, and therefore, the present value of lifetime earnings is the same for everyone. Although there is considerable cross-section earnings inequality, there is no lifetime inequality. While available data provide evidence that lifetime earnings do vary greatly across individuals, Becker’s model is too general to provide clear guidance concerning what portion of cross-section earnings inequality represents compensating differences on investment costs and what portion represents ability rents.

We need more specific models of human capital accumulation in order to precisely map talents and opportunities into life-cycle earnings profiles. The leading model of this kind is Ben-Porath’s (1967) analysis of human capital investments as an intertemporal optimization problem. Workers seek to maximize the discounted present value of their lifetime earnings by allocating their time between investments in human capital and

work and by choosing the optimal path of purchased investment goods over the life-cycle. The worker's problem is

$$\begin{aligned} \max_{s(\cdot), i(\cdot)} \int_0^T (w(v) - Pi(v))e^{-rv} dv \\ \text{s.t. } w(t) = R(1 - s(t))k(t), \quad k'(t) = \beta(s(t)k(t))^{\gamma_1} i(t)^{\gamma_2}, \end{aligned} \quad (4.4)$$

where $w(t)$ is the wage at t , $s(t)$ is the fraction of time spent training, $k(t)$ is the stock of human capital, $i(t)$ is purchased inputs used to produce human capital, R is the rental rate on human capital, P is price of purchased inputs, and r is the interest rate. $k(t)$ is "efficiency units" of human capital. Analysts typically assume that the equilibrium rental rate R and the price of purchased inputs P are constant over time. The expression for $k'(t)$ is the educational production function.

This is a standard control problem. The details of solution depend on the form of the production function. In this kind of model, the gross returns to using time either for direct production in the market or for investment in the production of human capital are increasing in the current stock of human capital. Ben-Porath's Cobb–Douglas form simplifies the problem immensely because it makes these two effects exactly offsetting and renders the optimal investment path of $k'(t)$ independent of $k(t)$. This property makes it possible to get closed form expressions for $w(t)$ and $k(t)$.

In the Ben-Porath framework, human capital $k(t)$ is used to produce both output and additional human capital. Therefore, one might treat the initial stock of human capital, $k(0)$, as an index of initial talent or ability. However, given this definition of ability, the Ben-Porath model does not yield the prediction that able workers invest more resources in human capital production. Both efficiency units of time investment, $s(t)k(t)$, and levels of purchased investment goods, $i(t)$, are independent of the current stock of human capital, $k(t)$. Workers with higher initial stocks of human capital have higher lifetime earnings, but they do not make larger investments in human capital production. Here, initial capital stocks are pure rents.

Becker's concept of ability is better captured by β , the scale parameter in Eq. (4.4). Consider two workers who begin their careers with identical human capital stocks, $k(0)$, but different values of β . Assume these workers make identical investments of purchased inputs, $i(0)$, and time, $s(0)$. The costs of these investments are the same in terms of forgone earnings and purchased inputs, but the high- β worker receives a greater return in terms of additional human capital, $k'(0)$. The worker with a higher β learns more efficiently, and it is easy to show that workers with more learning ability have higher lifetime disposable earnings. Further, for $t < T$, the more able worker invests more financial resources, $Pi(t)$, and more effective units of human capital, $s(t)k(t)$, in the production of human capital. Holding resource costs constant, workers who learn efficiently invest more in human capital and receive greater total returns from their investments. Thus, the

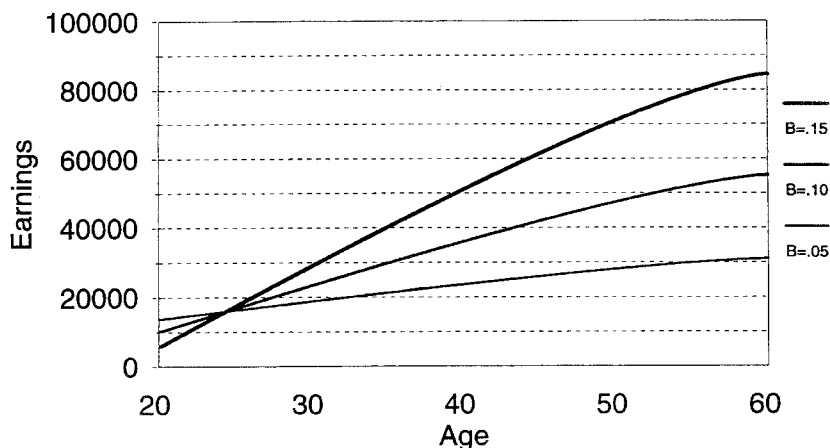


Fig. 4.2.

ability to learn appears to be closely related to Becker's notion of ability as a demand shifter.¹³

Lifetime accumulation of human capital also has implications for how the distribution of earnings for a particular cohort evolves over time. Figure 4.2 presents life-cycle earnings profiles for three workers in a Ben-Porath model with the same initial stocks of human capital but different learning efficiencies.¹⁴ The dispersion in earnings for a given cohort is greater for mature workers than for young workers. Further, the increase in cohort earnings dispersion begins at a point well into the workers' careers. In fact, in this example, dispersion decreases over the first few years and then increases dramatically after the more able workers, who make large initial investments in training, catch up.

¹³ The strong Markovian properties of Ben-Porath's functional specification afford so many analytical conveniences that its use has become routine. Alternatives are seldom considered. While convenient and simple, it is not necessarily descriptively accurate. For instance, the marginal costs of investment may be decreasing in $k(t)$. Then people who know more find it cheaper to invest. Initial stocks $k(0)$ as well as production function parameters affects optimum investment profiles and the effects of ability are more complex to describe. See Rosen (1976) for an example of this type. See Schultz (1975), Welch (1975) and Foster and Rosenzweig (1996) for extended discussion and evidence on how ability to learn affects productivity.

¹⁴ The three values for β are 0.05, 0.10, and 0.15. The other parameters are $\gamma_1 = 0.2$, $\gamma_2 = 0.075$, $r = 0.03$, and $R/P = 4$. The initial capital stock is normalized to one. The relative price of human capital roughly equals the ratio of mean annual earnings for full-year workers who are twenty years old, male, and high school graduates (\$16,300) to the mean cost of four quarters of college at a state university (\$4,021) for the year 1994. The earnings data come from the March, 1995 Current Population Surveys, and the tuition data come from The Digest of Educational Statistics (1996). There are no available estimates of β , γ_1 , and γ_2 . However, Haley (1973) does provide estimates from a model without purchased inputs. Here, the parameters are chosen in order to get initial investments of time and purchased resources that roughly correspond to three investment levels: full-time college, part-time college or vocational training, and a modest level of on the job training. See Von Weizacker (1993) for a complete exposition of similar models and similar simulation exercises concerning the evolution of the distribution of earnings for a cohort.

This U-shaped relationship between cohort earnings variance and cohort age is an important theme in the literature on human capital (Mincer, 1974). In Fig. 4.2, all workers begin their careers with the same initial productive capacity, the same financial constraints, but different abilities to learn new skills. Because the most able worker will make the largest investments in training, she actually begins her career earning less than the other two. Thus, the figure illustrates that the dispersion of earnings for a cohort may initially decrease with age as efficient learners begin to make fewer training investments and start to catch up to the earnings levels of less able workers. At some point, the earnings of able workers overtake the earnings of less able workers, and in this scenario, experience-earnings profiles in the cohort fan out and the dispersion of earnings increases dramatically thereafter.

When workers differ in their ability to acquire skills, the location of young workers in the distribution of current earnings for their cohort misstates their position in the distribution of lifetime earnings because greater levels of investment depress their current earnings relative to less able workers.¹⁵ Workers who learn more efficiently have both relatively steep earnings profiles and relatively high lifetime earnings. In contrast, among workers who differ only with respect to their initial stocks of human capital, $k(0)$, earnings profiles are parallel over the life-cycle,¹⁶ and a worker's position in the distribution of current earnings is a perfect indicator of position in the distribution of lifetime earnings (Lillard, 1977; Lillard and Willis, 1978; Weiss and Lillard, 1978; Weiss, 1986).

It is difficult to incorporate capital market imperfections in models of optimal human capital accumulation. Borrowing constraints require a utility based framework because they break down the separation between maximization of wealth and utility. But one can still represent some imperfections by exogenous differences in costs and returns for human capital investments. While "equal opportunity" corresponds to an environment where every worker can borrow and lend at the same interest rate, r , and every worker can purchase inputs for human capital production at the same price, P , an environment where agents face different levels of r and P generates "noncompeting groups" of workers. The presence of such groups affects not only the shape of the cross-section earnings distribution but also how the distribution of earnings for a given cohort evolves over time.

Here are some examples.

- (i) Government subsidies for both secondary and post-secondary education vary across countries and across regions within countries, and educational quality may differ among groups in the population. Both can be represented by interpersonal differences in P . Investment in human capital varies directly with R/P , the ratio of the rental rate on human capital to the price of purchased inputs in human capital

¹⁵ For example, medical students appear "poor" on current earnings accounts, but are wealthy on human capital accounts.

¹⁶ This result holds under the assumption that capital does not depreciate. In a model with skill depreciation, earnings profiles get closer together over the work life.

production. Persons facing low values of P invest heavily at the beginning of their careers and receive high earnings later in life.

- (ii) Optimal human capital investment models usually are based on the idea of mutually exclusive time allocations between work and training. However, there is an equivalent alternative set up where work and learning are jointly produced on the job. In this learning-by-doing framework, learning opportunities differ across jobs, and jobs pay different wages as an equalizing difference. Workers choose the sequence of jobs that implements their optimal investment program (Rosen, 1972). Labor market discrimination and other barriers restrict access of some people to jobs with greater learning opportunities and better chances for advancement. This is equivalent to an increase in the marginal costs of learning, represented by a decrease in β or an increase in P . Either way, those affected reduce their investments, and consequently, their experience-earnings profiles are flatter.
- (iii) Labor market discrimination can also cause rental rates (R) to differ across groups. This too reduces the marginal return to investment and restricts human capital accumulation incentives, again resulting in smaller lifetime earnings and flatter earnings profiles.¹⁷

4.3. Labor supply

Much earnings inequality is directly attributable to interpersonal differences in hours of work supplied to the market. This issue is important for empirical work and also bears on theory. The human capital models considered above treat labor supply as inelastic, independent of investment activities. However, one expects interactions because workers choose their investments and labor supply jointly. For instance, assuming that human capital investments do not affect the value of leisure, workers who expect to work more have greater incentives to invest in human capital because greater utilization of skills increases the rate of return to investment. In addition, those who invest more face higher costs of leisure at the margin and may substitute against nonmarket activities. Forces that determine the quantity and timing of leisure consumption can affect both the size distribution of earnings and the evolution of the distribution of earnings for a particular cohort.

Blinder (1974) was among the first to investigate these issues. He restricts his analyses to the pure effects of hours choices because he treats the path of wages as exogenous. Still, his work serves as a useful starting point. An agent's utility is a time- and goods-separable function of consumption and leisure. Workers borrow and lend at a fixed rate of interest, r , and face an exogenous life-cycle pattern of wages, $w(t)$. Each person

¹⁷ The natural extension of these models is to introduce explicit intergenerational linkages. These issues of intergenerational mobility are crucial to the study of economic equality and are treated separately by Piketty in this volume.

maximizes utility by choosing optimal paths of consumption $c(t)$ and leisure $l(t)$ given their initial wealth, $M(0)$, and a concave bequest function, $B[M(T)]$:

$$\begin{aligned} \max_{c(t), l(t)} \int_0^T e^{-\rho t} [U(c(t)) + V(l(t))] dt + B(M(T)) \\ \text{s.t. } c(t) + s(t) = rM(t) + w(t)(1 - l(t)), \\ M'(t) = s(t), \quad M(0) = M_0, \quad 0 \leq l(t) \leq 1 \end{aligned} \quad (4.5)$$

where $s(t)$ is savings at t , and ρ is the rate of time preference. To illustrate the solution assume $U(c(t)) = \log(c(t))$, and $V(l(t)) = \varphi \log(l(t))$ and $\rho = r$ for all workers. The first order condition for $l(t)$ is

$$l(t) = \frac{\varphi c(t)}{w(t)}. \quad (4.6)$$

When $\rho = r$, $c(t) = c(0)$ is a constant, and earnings are given by

$$E(t) = w(t) - \varphi c(0), \quad (4.7)$$

so actual earnings profiles, $E(t)$, and potential earnings profiles, $w(t)$ —what the worker would earn if working full-time—are parallel. Because consumption $c(0)$ is an increasing function of lifetime earning potential, the gap between lifetime earnings and lifetime earning potential increases with earning potential. Thus, when the path of potential earnings is exogenous and leisure is a normal good, the consumption of leisure mitigates lifetime earnings inequality. Blinder (1974) demonstrates a similar result concerning overall income inequality. Workers who begin life with large initial asset endowments, M_0 , consume more leisure over their lifetime. Nonearned income is a substitute for earnings.

If all workers have the same tastes, the positive relationship between wealth and consumption of leisure ameliorates overall income inequality due to individual differences in asset income and earning potential. However, all workers do not have the same tastes, and tastes for leisure may well be correlated with individual wage rates and assets. Blinder uses simulations to gauge what sources of individual heterogeneity contribute significantly to the observed inequality in labor incomes. His results suggest that individual differences in wage profiles are the most important factor in determining earnings inequality, but there is also an interaction between tastes and wages. If wages and preferences for leisure are negatively correlated, realized lifetime earnings inequality may be much greater than the underlying inequality in potential earnings. This finding is important because we expect tastes to influence earnings capacities. In Blinder's framework, correlations between potential earnings and tastes are simply correlations between worker endowments. However, in human capital models, earnings capacity is,

in part, determined by tastes. Because workers with weak preferences for leisure forfeit relatively little in utility terms when they work, they have additional incentive to invest in market skills. In short, we expect future earnings capacity to vary inversely with current tastes for leisure.

Adding human capital investment to problem (4.5) yields a more complete model that analyses three uses of time: investment, $s(t)$, leisure, $l(t)$, and work time $(1 - s(t) - l(t))$.¹⁸ We maintain the Ben-Porath technology for human capital investment in Eq. (4.4), but ignore bequests and purchased inputs, and write the instantaneous utility function as $u(c(t), l(t))$. Earnings capacities evolve as workers make investments in human capital. Assuming that the rate of time-preference equals the rate of interest, first order conditions for $c(t)$, $l(t)$ and $k(t)$ satisfy

$$u_c(c(t), l(t)) = \lambda \quad (4.8)$$

$$u_l(c(t), l(t)) = \lambda Rk(t), \quad (4.9)$$

$$\frac{R[s(t)k(t)]^{1-\gamma}}{\beta\gamma} = \int_t^T R(1 - l(\tau)) \exp(-r(\tau - t)) d\tau. \quad (4.10)$$

λ is the multiplier associated with the lifetime budget constraint. The right hand side of Eq. (4.10) is the discounted marginal value of a unit of current investment. The left-hand side is the marginal cost of additional human capital, $k'(t)$, in terms of forgone current earnings.

Equations (4.8) and (4.9) are the standard intertemporal substitution conditions, with earning capacity, $Rk(t)$, replacing the wage rate. It is clear from Eq. (4.9) that the marginal utility of leisure is inversely proportional to earnings capacity along the optimum path (over the life cycle). If the marginal utility of leisure is independent of current consumption, this implies that leisure varies inversely with earnings capacity over the life cycle. As in the standard Ben-Porath model, time spent investing, $s(t)$, declines monotonically with age. Thus, when we examine a cross-section of workers, part of the observed difference in earnings between experienced and inexperienced workers may reflect the fact that experienced workers not only enjoy higher earnings capacity because of previous human capital investments but also spend less time investing in human capital and consuming leisure. A similar result holds in Blinder's framework if one assumes that earnings capacity rises over time, but human capital models generate rising earnings capacities endogenously.

Further, while the model yields few general comparative dynamics results, it does provide a framework for discussing how differences in tastes for leisure across individuals may contribute to differences in the evolution of life-cycle earnings profiles. Take the special case where preferences are separable over both time and goods, and

¹⁸ Examples include Heckman (1976), Ryder et al. (1976), and Blinder and Weiss (1976). Weiss (1986) provides a survey of life-cycle human capital models.

assume interior solutions for all $t < T$. Now, consider two workers with identical skill endowments but different tastes for leisure. Because worker A has a stronger preference for leisure, he begins his career by devoting more time to leisure than worker B. Worker B always invests more in terms of efficiency units of human capital, $s(t)k(t)$, because his weak tastes for leisure increase the returns from investments in market human capital. Therefore, over time, the earning capacity of B grows apart from the earning capacity of A. Since B always has higher earning capacity and weaker tastes for leisure, B always consumes less leisure than A.

Ignoring corner solutions, A always consumes more leisure than B. But, one cannot say that A always works less because A may spend less time training. As the workers age, differences in leisure consumption become more pronounced because earnings capacities continue to grow apart. However, differences in training must eventually diminish because training tends to zero for both workers as $t \rightarrow T$. Thus, differences in tastes for leisure lead to life-cycle patterns of leisure and training that eventually magnify earnings differences among experienced workers. Experienced workers with weak tastes for leisure have relatively high earnings capacities because of previous investments. For experienced workers, the combination of weak tastes for leisure and high earnings capacities reduces current leisure consumption and, at some point, must translate into both higher earnings and more labor supply.

While the exact life-cycle relationship between tastes for leisure and labor supply is ambiguous in general, it is clear that, holding ability constant, those with weaker tastes for leisure have higher discounted lifetime earnings.¹⁹ Further, human capital models demonstrate that this result is more than a link between labor supply and tastes for leisure. There are two components to the link between tastes and earnings. Given earnings capacity, workers with weak tastes for leisure work more. But earnings capacity is not independent of tastes. Weak tastes for leisure enhance investment and therefore raise earnings capacity.

Adding labor supply considerations to human capital investment models extends the economic implications of these models. People with either weaker tastes for leisure (inferior opportunities for work outside the market) or lower costs of human capital production invest more. Thus economic development, decreasing fertility, and other factors affecting the labor force participation of women increase the demand for human capital investments and augment the future market supply of skill. What remains for future research is a better understanding of the demand for skilled labor and how such shifts in the supply of skill affect the equilibrium rental rate on human capital.

¹⁹ If two workers have the same opportunities, the one with weaker tastes for leisure will enjoy more goods consumption. This requires higher lifetime earnings.

5. Insurance, agency and earnings

If firms could pay workers strictly on the basis of their own production, most incentive and labor relations problems between employers and employees would disappear. Pay would be contingent on a mutually acceptable outcome. Few labor contracts work this way because individual output is costly to assess at each point in time. Payments of salaries and wage rates for a worker's effort and time are the norm, and firms loosely structure the detailed duties and obligations of employees. Since workers typically have some degree of control and discretion over their work, they have opportunities to pursue their own interests instead of their employers' interests. Examples are shirking and taking malfeasant actions, such as theft of goods from the firm and accepting payments from outside contractors.

The economic theory of agency analyzes such problems by describing internal compensation mechanisms that resolve conflicts and align the interests of employer and employee. The literature on agency examines the scope and limitations of different payment schemes, such as piece rates, for eliciting desired worker behavior, but our interest in agency theory is restricted to its role in understanding the distribution of compensation in labor markets. In this regard, agency models provide an important contrast to the models reviewed earlier in this survey. In most theories of earnings distributions, competition ensures that individual earnings reflect individual productivity, and the chief aim of theory is to explain the distribution of individual productivity. The agency literature reverses things: methods of compensation affect the allocation of worker effort across tasks and therefore determine the distribution of individual productivity. Moreover, observed wage rates at any point in time may not reflect personal productivity at that point.

A related literature focuses on a different information problem. When firms can observe productivity, risk averse workers who are not privately informed of their personal productivity do not want to be paid strictly according to their *ex post* output. They want insurance against random factors that affect productivity but are beyond their control. This demand for insurance shapes compensation policies by weakening the links between pay and *ex post* performance.

Below we discuss implications of both agency models and models of earnings insurance for the distribution of earnings. We begin by describing the insurance components of wage payments. This is the starkest possible departure between personal pay and personal productivity: pooling risks introduces voluntary redistributions of income that tend to compress wage distributions. Next, we consider agency models where firms use policies that increase wage dispersion in order to elicit effort. Tournament models of competition for promotion and models of bonding and deferred compensation loosen the links between current productivity and current earnings.

We conclude with an examination of efficiency wage models that address monitoring problems. In these models, firms use wage premiums to elicit certain desired aspects of employee performance, but other nonwage aspects of pay, such as performance bonds are not present and labor markets do not clear. These models do not have clear implica-

tions for the overall distribution of earnings, but they do provide a possible explanation for observed differences in mean wages across sectors of the economy.

5.1. Attitudes toward risk

Worker preferences alone can give rise to voluntary redistribution in the presence of uncertainty. Friedman (1953) is the first to address earnings inequality in such terms (see also Kanbur, 1979). Risk averse workers have incentives to equalize their incomes when earnings have stochastic components. If ex post outcomes contain random elements and no one has private information, a group of ex ante identical risk averse workers who know the probability distribution of outcomes pool risks and insure each other. All agree ex ante to a binding contract in which everyone contributes their ex post earnings to a common fund. All receive the ex post mean, and ex post income is completely equal even though the distribution of ex post productivity might be dispersed.²⁰

Full ability-insurance is not widely observed in the labor market as a whole due to moral hazard and adverse selection. People who turn out to be successful would be reluctant to share their windfalls with the less fortunate, and if they did, they would have little incentive to develop and exploit their skills. Further, people who thought themselves more talented ex ante would be reluctant to participate in such a scheme in the first instance.²¹

Harris and Holmstrom (1982) provide an interesting model of earnings insurance that is an elaboration of the simple learning model described in Section 3.1. The record of production gradually reveals a person's productivity, and there is market competition for workers. If workers and firms could write lifetime employment contracts, risk neutral firms could provide complete earnings insurance for workers. However, in the absence of such contracts, labor mobility limits the ability of firms to provide insurance. Ex post, highly productive workers have an incentive to leave a firm that pays each worker according to the ex ante expected productivity of all its workers. Nonetheless, even in an environment where workers are free to change firms, an employer can provide partial insurance by matching outside offers and guaranteeing that a worker's earnings never

²⁰ Friedman himself investigated the possibility that this kind of insurance is not observed because people are risk-loving rather than risk averse. Workers with Friedman-Savage preferences (risk-loving in a certain range) have incentives to increase the dispersion of income by gambling. Recently Bergstrom (1986) and others (see Freeman, 1996 and references therein) have shown that Friedman-Savage risk preferences occur when the marginal utility of material goods consumption interacts with the type of work a person does or with other indivisible life-circumstances, such as place of residence. However, these types of risk preferences need not be addressed by labor market mechanisms. Purely monetary side gambles are sufficient. After winners and losers settle these side bets, they can make their occupational and other indivisible choices deterministically (Rosen, 1996), just like the models in Sections 1-4.

²¹ There is a certain awkwardness in the pure theory of insurance as applied to labor market contracts. Why should the arrangements flow through firms rather than being managed by third party insurance companies? One possibility is that the firm is better able to control the worker's work performance, on which the insurance is contingent.

fall below his starting wage, no matter what the subsequent record reveals about personal productivity.

Since the firm rationally expects to pay higher wages to successful workers later on, the starting wage must be less than the unconditional mean productivity in the group. But risk averse workers are willing to accept a lower starting wage for the privilege of being insured against subsequent wage cuts should their productivity record prove unfavorable. The difference between the guaranteed starting wage and mean productivity of the entry cohort represents an up-front insurance premium. It is paid by workers and is necessary to finance the insurance. Likewise, when a worker who performs well receives a new wage offer, the new offer is less than the worker's expected productivity given the new information. The worker's ability is still uncertain, and the worker is willing to accept a new wage somewhat below expected productivity in exchange for a guarantee that the new wage will never fall in the future. As the worker becomes more experienced, there is less uncertainty about his true ability, and it costs firms less to provide a guarantee against future wage cuts. Therefore, as a cohort of workers gains experience, those receiving wage increases pay smaller premiums for earnings insurance.

The predictions of this model are similar, but stronger than those of the sorting model in Section 3.2. In the sorting model, wages become more dispersed as the market learns more about individual abilities. Over time, average wages rise and the distribution of wages for a cohort spreads asymmetrically. Those who learn that they possess lower levels of ability mitigate their losses by switching to jobs where output is less sensitive to ability, while those who learn the opposite magnify their gains by switching to jobs that more fully exploit their talent. In the Harris and Holmstrom framework, learning also creates wage growth and asymmetric wage dispersion because the distribution is always truncated at the starting wage. But in addition, wages of individuals never fall, as they would in the pure sorting model. There is a kind of "downward wage rigidity".

In general, worker demands for insurance compress wages relative to the distribution of individual productivities, but, because workers and firms cannot write complete insurance contracts, the resulting compression is one-sided. Next, we turn to models of effort elicitation. These models also generate asymmetric wage distributions, but here the emphasis is on stretching the right tail of the distribution not truncating the left tail.

5.2. *Effort incentives and compensation in internal labor markets*

Wages differ substantially across job titles within firms. Not only are there substantial pay distinctions between rank and grade, but empirical work shows that wage jumps associated with promotions account for a high proportion of a worker's wage growth within firms, especially in white-collar jobs (Baker et al., 1994). At first blush, this is hard to reconcile with standard productivity theory. Though a person's job activities usually change after a promotion, one's inherent skills and productivity do not jump on promotion day. Evidently, firms tie wages to jobs as well as people.

Job transfers of existing personnel and promotions from within can be thought of as an internal labor market. A military organization, where outside recruitment occurs only at the lowest ranks and all higher level positions are filled by internal promotions, is the most familiar example. Other organizations recruit at all levels but fill most vacant positions through internal transfers. Private law firms promote associates to partnership status, academic departments grant tenure and change the rank of professors, and large firms draw many of their executive staff from lower ranking positions in the same firm.

The indivisibility of job assignments is the main reason for internal recruitment of personnel: there can only be one chief executive officer at a time. Promotion policies allow firms to learn worker capacities for more important jobs by observing performance on less important ones; and the workers gain specific knowledge of the firm's operations and procedures that enhance future productivity in such positions. Issues of testing and sorting naturally arise in the assignment of persons to positions in internal labor markets. Firms promote workers who surpass a certain threshold of performance. Raises associated with promotions provide incentives for performance, and holding the firm's wage structure constant, the levels of various performance thresholds also affect incentives.

Let x be a worker's productivity on the current job, with $x = \mu + \epsilon$, where μ is the worker's effort and ϵ is a random variable drawn with mean zero.²² $C(\mu)$ is the cost to the worker associated with effort level μ . $C'(0) = 0$ and $C''(0) > 0$ for all μ . Normalizing the price of x at unity, the efficient choice of μ for a risk neutral worker occurs when the marginal cost of effort equals its marginal return, or $C'(\mu) = 1$. The probability of passing the promotion standard²³ is calculated from the distribution of ϵ as $p = p(\mu, s)$, where s is the pass threshold measured in units of x , and $p_\mu > 0$ and $p_s < 0$. The firm promotes those who pass to a job that pays wage W_1 . Those who fail go to a lower paying job W_2 .²⁴

The expected return to the worker is

$$p(\mu; s)(W_1 - W_2) + W_2 - C(\mu). \quad (5.1)$$

The worker puts forth effort to satisfy the first order condition

$$p_\mu(\mu; s)(W_1 - W_2) = C'(\mu). \quad (5.2)$$

Comparative statics imply that μ is increasing in both $(W_1 - W_2)$ and s , when s is small. Therefore, combinations of the promotion bonus, $(W_1 - W_2)$, and the height of the bar, s ,

²² We also assume that the density, $g(\epsilon)$, is single-peaked at 0.

²³ If the cumulative distribution of ϵ is $G(\epsilon)$ and the worker is promoted if $x > s$ then $p(\mu; s) = 1 - G(s - \mu)$. Note that this is not necessarily concave in μ so first order conditions must be carefully checked for optimality. These complexities are ignored here.

²⁴ There is no mobility here. Workers and firms agree on a lifetime employment contract that gives the workers their reservation level of utility. If workers are free to leave in the second period, firms face additional participation constraints in choosing their wage structure.

can be found that achieve the efficiency condition $C'(\mu) = 1$. Firms pay increasing and convex costs to set their standards. With homogenous firms and workers, competition by firms for workers ensures an equilibrium combination $[(W_1 - W_2), s]$ that not only elicits efficient effort but also ensures that, given efficient effort, workers receive their opportunity cost and firms earn zero profits.²⁵

$$pW_1 + (1 - p)W_2 = E(x) = \mu. \quad (5.3)$$

It is interesting that although the average wage equals average productivity, the actual wage almost never equals ex post realized productivity. For instance, the probability that a person's realized value of x equals either W_1 or W_2 is zero, if ϵ is continuous. Here, wage dispersion does not reflect ex post differences in personal productivity. Rather, wage dispersion elicits ex post productivity.

In many instances, it may be easy to rank workers on an ordinal performance scale but difficult or impossible for firms to rank workers on a cardinal scale. Even when it is clear that worker A is better than worker B, firms may not know how much better. Under these circumstances, firms make promotion decisions based on relative performance, such as head to head competition among lower ranking workers vying for higher level positions. The noise introduced by common-error components in measuring individual performance can be eliminated by comparing contestants with each other. The result is as a "tournament" (Lazear and Rosen, 1981; Nalebuff and Stiglitz, 1983; Green and Stokey, 1983; McLaughlin, 1988) or some other form of relative performance evaluation (Holmstrom, 1982).

In tournament theory, contestants vie against each other rather than against a fixed standard. This brings strategic considerations into the problem above, because the probability of promotion for each player depends on what others are doing. The efforts of all other contenders replaces s in the probability of winning function, and we must analyze wages as outcomes of a noncooperative game. Nonetheless, prizes exist that produce the socially efficient level of effort among ex ante homogeneous players.

The tournament literature provides other interesting results concerning the distribution of earnings within firms. Since more contestants competing for a fixed number of promotions adversely affects incentives (O'Keeffe et al., 1984), the age and experience distributions within a firm affect its internal wage policy. Further, analysis of sequential contests in promotion ladders reveals another reason why compensation is extraordinarily high in top level positions. In tournaments where contestants are eliminated from further contention after failing, the difference in rewards between adjacent ranks provides only part of the incentives to perform. There is substantial option value in the probability of contending for even larger prizes in better jobs down the line. But in the last rounds of the tournament, that option plays out. Maintaining performance incentives at this crucial stage requires that the option value be replaced by an extraordinary jump

²⁵ Although regions exist where effort declines with increases in the standard given $(W_1 - W_2)$, firms never set standards this high.

in wages at the top of the organizational chart (Rosen, 1986). Again, wages do not necessarily equal ex post personal productivity of persons who win such prizes but do affect the productivity of others in the organization.

5.3. Agency and performance bonds

Contests or tournaments constitute incentive schemes that magnify earnings dispersion within a firm for a particular cohort of workers. A given cohort of workers enters a contest and over time, their earnings grow apart as some win and some lose. However, a different class of agency models addresses earnings inequality across cohorts. Models of this type highlight the role of performance bonds and deferred compensation as monitoring devices.

The basic solution to agency problems is a performance bond, in which a worker is put in a position to lose something if shirking or malfeasance are detected by the firm. The threat of dismissal alone does not provide sufficient incentives if an alternative job with the same pay (utility) can be readily found, and paying wages in excess of opportunity costs just to prevent shirking essentially creates rents for workers fortunate enough to land such jobs. In agency theory proper, other components of pay are adjusted to eliminate these rents and achieve market clearing. These take the form of payments at the front (bonds) and back (pensions) of the contract. Section 5.4 discusses efficiency-wage models, where these extra market-clearing adjustments are not admissible.

Workers in positions to take malfeasant actions against the firm can be deterred by sufficient monitoring (Alchian and Demsetz, 1972), but, like police and lawyers, monitoring is expensive. A self-enforcing payment mechanism that induces workers not to take advantage of the firm may be less expensive and achieve the same result. Obviously this requires penalizing malfeasant behavior (the “stick” approach) or rewarding good behavior (the “carrot” approach), so that workers find it in their best self interest to behave as the firm desires them to behave (Lazear, 1995). Ross (1973) and Becker and Stigler (1974) established that an up-front performance bond and back loaded pension accomplish this.

Following Becker and Stigler (1974), consider the last period of job which pays a wage W_n and offers a chance at malfeasance worth x to the worker. Knowing that malfeasance can occur, the firm installs an imperfect monitor that detects malfeasance with probability p . The firm fires the worker if it detects malfeasance, and the worker may take a job elsewhere that pays V_n . The worker’s expected return to malfeasance is $(1 - p)(W_n + x) + pV_n$. The return to honesty is W_n itself. Equating the two gives the minimum honesty-inducing wage

$$W_n = V_n + [(1 - p)/p]x. \quad (5.4)$$

Honesty requires paying a wage premium of $[(1 - p)/p]x$ that varies inversely with the probability of being caught.

Consider next a two-period job in which the wage paid in the second (last) period is given by Eq. (5.4), and thus, workers do not shirk in the second period. What wage is required to deter shirking in the first period? The present value of earnings in a job where the worker is deterred from malfeasance in both periods is $W_{n-1} + W_n/(1+r)$, where r is the discount rate. If the firm dismisses all malfeasant workers it detects in the first period, the expected present value of malfeasance is $(1-p)[W_{n-1} + x + W_n/(1+r)] + p[V_{n-1} + V_n/(1+r)]$. Equating present values gives the minimum payment needed to deter malfeasance in the first period as

$$W_{n-1} = V_{n-1} + [(1-p)/p][r/(1+r)]x. \quad (5.5)$$

Equation (5.5) generalizes to all contracts lasting an arbitrary number of periods.

Notice the difference between Eqs. (5.4) and (5.5). The wage premium needed to deter malfeasance is largest in the final period of the contract because the threat of termination has less deterrent value the shorter the remaining horizon. But, in every period of the contract, $W_t > V_t$. Since there are rents in every period, the expected present value of the job strictly exceeds that of the alternative. The difference in present values works out to be $[(1-p)/p]x$. To equalize present values and avoid job rationing, workers are obliged to post front money of $[(1-p)/p]x$. Shirking may be eliminated without job rationing under the following contract: (i) workers post a performance bond before work begins, (ii) wages at each period equal opportunity wages plus interest on the bond, and (iii) deferred pay at the end of the contract equals the bond itself.

There is an important general point in this example. Current wage rates alone are insufficient to achieve efficient incentive alignment in many agency problems. In this example, front and back-loaded payments, as well as wages are needed. In a sense, one price, the wage rate, has to do too many tasks (Lazear, 1995). The firm needs to attract applicants for the job and provide performance incentives once workers take the job. Generally more than one price is needed to accomplish more than one economic function. The bond creates a "price", as it were, for malfeasance.

Pensions and deferred pay represent significant components of total compensation in many jobs where malfeasance is of obvious concern. Examples include the military, police, judges, politicians and some civil servants. Of course pensions are of growing importance in total compensation for other reasons having to do with the tax advantages of saving for retirement in this way. Nonetheless, most private pension plans that are based on a person's previous wage record, greatly penalize departures from the firm prior to the standard retirement age and serve a deterrence function in this respect. The data do not support the up-front bonding aspects of the solution. Explicit performance bonds are seldom observed in labor markets. If x is large and p is small, large bonds might be necessary but infeasible for liquidity constrained workers. Further, workers would be reluctant to post performance bonds when firms have the opportunity to falsely claim malfeasance by workers and seize their bonds.

Lazear (1979) developed an important extension of the basic model by noting that deterrence does not require strict equality between the values of malfeasance and non-

esty in each period. Inequalities also work. In Lazear's model there is no explicit up front bond. Instead workers gradually post the equivalent of a bond by working for less than their full productivity to the firm when they are young. The firm breaks even because it pays them more than their direct productivity when they are older. In a sense, workers are building an implicit "equity" or "partnership" position in the early years and receiving "stakeholder" returns in the form of wage premiums and pensions in later years. In this model, wage-experience profiles are upward sloping for incentive reasons. The distribution of wages within a firm is more dispersed than it would be in an environment without monitoring concerns.²⁶

Akerlof and Katz (1989) point out that the implicit bonding scheme proposed by Lazear does not work at the beginning of careers. At each point in time, the worker weighs the benefits of shirking against its expected cost, which equals the value of the implicit bond that the worker has already posted times the probability of detection. At the beginning of the employment relationship, the value of the implicit bond is zero, and the worker has nothing to lose by shirking. Implicit bonding by workers requires some mechanism that gets around this "start up" problem. For instance, the firm may offer entry level jobs where output is easily observed, with workers posting bond by being paid piece-rates below their marginal product. After posting the bond, the worker can then be promoted to more complex jobs where output and effort are not directly observed. For workers who begin their careers with few skills, minimum wage laws or other employment regulations may frustrate attempts to post bonds in this manner. Nonetheless, there are other degrees of freedom. Firms may engage in intensive direct monitoring until workers develop the human capital necessary to post a bond while working at the minimum wage. If bonds cannot be posted implicitly, upward sloping wage profiles still serve as an incentive device, but the market does not clear. Efficiency wage models explore these issues.

5.4. *Efficiency wage models*

Efficiency wage models come in several flavors, but most share the common feature that workers are not allowed to post bonds. We devote our attention to the shirking model.²⁷ The model of Bulow and Summers (1986) uses inter-industry differences in monitoring technologies to explain how wage levels of observationally similar workers

²⁶ In addition, the model provides a rationale for mandatory retirement. When the contract reaches a termination date set at the beginning of the contract, the wage being paid at that time is greater than the worker's opportunity cost, and the worker "prefers" to keep working. The firm must terminate the worker by mandatory retirement at that point else the ex ante scheme would go insolvent. This is one of the only economic theories of mandatory retirement that has been proposed.

²⁷ Shapiro and Stiglitz (1985) also provide a well known shirking model. Some efficiency wage models address asymmetric information about worker quality, turnover costs or the nutritional needs of workers in undeveloped countries. In all these cases, up front payments are not allowed and ex ante rents and nonprice rationing of jobs creates unemployment. Bonds are not so crucial in efficiency wage models based on the sociological concept of voluntary gift exchange.

differ across sectors of the economy in a cross-section. Firms in the “primary” sector are, by assumption, unable to effectively monitor their workers. Direct expenditures on supervision do not change the probability of catching a worker who is shirking. Firms in the “secondary” sector are able to monitor their workers perfectly at zero cost.

In the primary sector, firms catch workers who shirk with probability p . Primary sector firms cannot change p , and by assumption, workers cannot post performance bonds either implicitly or explicitly. Firms may fire shirkers if they catch them shirking, but firms have only one additional means to deter shirking. They can make dismissal more costly by paying wages above the going wage rate in the secondary sector.²⁸ The wage premium required to prevent shirking is an increasing function of total employment in the primary sector because a large primary sector makes it relatively easy for a dismissed worker to find a new primary sector job. This no-shirk function is depicted in Fig. 5.1. The intersection between it and the demand for primary sector workers determines sectoral employment and the wage premium enjoyed by primary sector workers. The model generates an equilibrium where identical workers in different sectors of the economy earn different wages. The relative demand for primary sector output determines both primary sector employment and the magnitude of the inter-sector wage gap.

Wage levels differ considerably across the industries of modern economies. Controlling for observed differences among workers in various industries eliminates some of the interindustry wage variability, but much remains.²⁹ This theory offers an explanation for the variance remaining after standardizing for measured skills. It is easy to imagine a generalization of the model where w^* represents the value of home production, and there are N no shirk conditions that correspond to N different industries with various degrees of monitoring problems. Such a model would generate a pattern of differences in wages across industries that reflects differences in monitoring technologies across various workplaces rather than inherent differences in worker skill. However, as an empirical matter, unobserved differences in skills of workers across industries appear to account for at least part of these wage differentials.³⁰

Efficiency wage models of shirking problems generate a distribution of wages across sectors of the economy that reflects heterogeneity across firms with respect to monitoring technologies. In these models, workers are not allowed to post bonds, and because this option is shut down, firms use wage premiums as a mechanism to provide performance incentives. Complete market agency models involve either implicit or explicit performance bonds, and over a worker's entire tenure with a firm, the present value of his lifetime wages equals his opportunity cost. It is difficult to separate these models empirically because available data provide little information about monitoring technologies.

²⁸ The result is similar to the one presented above in our discussion of Becker and Stigler (1974). However, the Bulow and Summers model is static and does not address how time horizons interact with the threat of dismissal.

²⁹ See Krueger and Summers (1988).

³⁰ See Gibbons et al. (1997), Gibbons and Katz (1992), and Murphy and Topel (1990).

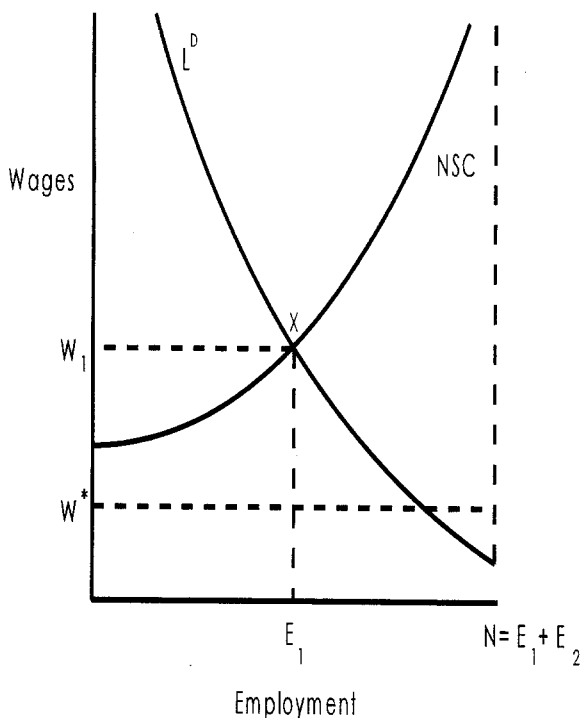


Fig. 5.1.

However, previous studies have not demonstrated a link between monitoring practices and inter- industry wage premiums.³¹

6. Conclusion

We began by noting that earnings distributions are always skewed. The right tail is much longer than the left. Selection and sorting models illustrate that when workers choose the job that best suits their skills, observed earnings do not include many of the worst outcomes from distributions of potential earnings associated with specific jobs. Insurance models describe how partial insurance contracts truncate wage distributions on the left, while tournament models illustrate how firms stretch the right tail in order to provide incentives. Human capital models can generate skewed distributions because individual endowments like learning ability, productive capacity and tastes for leisure can interact to generate considerable heterogeneity in investment behavior.

³¹ See Neal (1993) and Leonard (1987).

Human capital models stand out because they tie observed differences in earnings across education and occupation groups to observed differences in skill investment. Human capital models also provide unique insights concerning how the distribution of earnings evolves for a particular cohort. Basic learning models as well as sorting models predict monotonically increasing earnings dispersion for a given cohort as the cohort ages. In human capital models, earnings dispersion can actually decrease with cohort age at the beginning of the life-cycle before increasing dramatically during middle age. When workers have the same initial productive capacities but different abilities to learn, earnings capacities grow steadily apart over the life-cycle even though observed earnings levels do not fully reflect this divergence in the early years. During later working years, investments are minimal, and earnings are closely tied to earnings capacities. This yields striking dispersion because capacities have grown apart for years.³²

However, human capital models cannot generate the long right tails that we observe in modern earnings distributions. For instance, we cannot easily explain the salaries of top chief executives simply as the product of an exceptional talent for learning new skills. Rather, such earnings more likely reflect the optimal assignment of rare talents to a specific set of tasks. Models of scale economies and the span of control tell us why the persons assigned to such jobs earn such enormous salaries.

The literature on scale economies and span of control describes a link between technology and the distribution of earnings. Because technology determines the personal scale of a worker's operations, it affects the demand for skill and the distribution of ability rents. To the extent that recent advances in information technology have made it easier for talented individuals to have larger spans of control, the information revolution may be a driving force behind the recent explosion of earnings in the very top percentiles of the distribution. This feature of span of control models is quite unique among the models discussed in this survey. Most of the other models presented here generate earnings distributions entirely from supply decisions. They illustrate how the allocation of talents, time and energy generate and sustain earnings inequality. Selection models illustrate how workers supply their endowments to tasks that generate the highest earnings. Human capital models describe skill formation and demonstrate that inequality is necessary when different types of work require different levels of skill investment. Models of the investment process illustrate how talent and financial resources affect investment decisions and therefore demonstrate how human capital accumulation links initial endowments, family background, and realized inequality. Agency models focus on the supply of effort and describe wage inequality within firms as an incentive device.

³² Most of the empirical work on this topic provides evidence on the evolution of the variance of log earnings for a particular cohort. Throughout the 1970s and early 1980s, several studies found evidence of a U-shaped relationship between the variance of log earnings and cohort work experience. Mincer (1974), Smith and Welch (1979), Hause (1980), and Dooley and Gottschalk (1984) address this topic. The U shaped relationship in log variances implies that the rate of increase in the dispersion of earnings levels is greatest during the prime working years and that the variance of earnings (measured in levels) is much greater during prime working years than during the initial years of a cohort's career.

None of these models say much about the determinants of labor demand or the elasticities of demands for different types of skill. This may be an important omission because much recent research explores potential “demand driven” explanations for the recent rise in wage inequality observed in many Western countries. A large empirical literature debates competing claims about the nature of labor demand and its response to changes in technology or trade barriers. We do not explore these issues here,³³ but we do note that most theoretical work on the distribution of labor earnings does not analyze the forces that shift or shape the demands for various types of market skills.

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³³ Because the models described here do not address the elasticities of demand for various types of workers, they say little about how demographic changes affect the wage structure through changes in the relative supplies of various skills. Murphy and Welch (1992) and Katz and Murphy (1992) provide evidence concerning the relative importance of shifts in the demand and supply at different levels of worker skill.

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